



OPEN ACCESS

EDITED BY

Vishnu D. Rajput,
Southern Federal University, Russia

REVIEWED BY

Adnane Beniaich,
Mohammed VI Polytechnic University,
Morocco
Fakher Abbas,
Chinese Academy of Sciences (CAS),
China

*CORRESPONDENCE

Waqar Akhter Ansari
✉ waqar.ansari@gmail.com,
✉ waqar.ansari@
marwadieducation.edu.in

RECEIVED 15 January 2026

REVISED 08 April 2026

ACCEPTED 13 April 2026

PUBLISHED 13 May 2026

CITATION

Zhapparbergenov R, Tautenov I,
Bekzhanov S, Khakberdiev O, Aamir M
and Ansari WA (2026) Phosphogypsum
application in agriculture: a review of soil
fertility, plant nutrition, and
environmental effects.
Front. Soil Sci. 6:1788339.
doi: 10.3389/fsoil.2026.1788339

COPYRIGHT

© 2026 Zhapparbergenov, Tautenov,
Bekzhanov, Khakberdiev, Aamir and
Ansari. This is an open-access article
distributed under the terms of the
[Creative Commons Attribution License](https://creativecommons.org/licenses/by/4.0/)
(CC BY). The use, distribution or
reproduction in other forums is
permitted, provided the original
author(s) and the copyright owner(s) are
credited and that the original publication
in this journal is cited, in accordance
with accepted academic practice. No
use, distribution or reproduction is
permitted which does not comply with
these terms.

Phosphogypsum application in agriculture: a review of soil fertility, plant nutrition, and environmental effects

Rakhmetulla Zhapparbergenov¹, Ibadulla Tautenov¹,
Serik Bekzhanov¹, Obid Khakberdiev², Mohd Aamir³
and Waqar Akhter Ansari^{4*}

¹Department of Agrarian Technology, Korkyt Ata Kyzylorda University, Kyzylorda, Kazakhstan,

²Department of Soil Science and Farming, Tashkent Institute of Irrigation and Agricultural

Mechanization Engineers National Research University, Tashkent, Uzbekistan, ³Department of Life Sciences, Parul Institute of Applied Sciences, Parul University, Waghodia, Gujarat, India, ⁴Marwadi University Research Centre, Department of Agriculture, Faculty of Science, Marwadi University, Rajkot, Gujarat, India

Phosphogypsum (PG), a calcium sulfate dihydrate by-product of wet-process phosphoric acid production, is generated globally at over 300 million tons per year, yet only 14% is currently reused or recycled, with the remainder stockpiled or discharged into coastal areas. Its agricultural use as a soil amendment has grown considerably, yet a rigorous, integrated synthesis of its agronomic benefits, environmental risks, and practical limitations has been lacking. This review addresses that gap through analyzing questions under what conditions, at what rates, and with what co-amendments can PG be applied safely and effectively to degraded agricultural soils? Literature was searched across PubMed, Scopus, Web of Science, and Google Scholar using Boolean keyword combinations centered on PG and soil properties, crop yield, radionuclides, heavy metals, soil microbiology, and circular economy themes, covering 2000 to 2025. The review demonstrates that PG improves soil aggregate stability, reduces bulk density, increases porosity and water permeability, and corrects sodicity and secondary salinization through Ca^{2+} – Na^{+} exchange in saline, sodic, and magnesium-affected soils. It reliably supplies calcium and sulfate–sulfur across vegetable, cereal, legume, and industrial crop systems, and alleviates subsoil aluminum toxicity in acid tropical soils where lime is ineffective. The novelty of this review lies in integrating agronomic performance, microbial community responses, and radionuclide and heavy metal risk into a single analytical framework, positioned within a circular economy context. Critical knowledge gaps identified include insufficient long-term multi-year field data, geographically restricted study populations, and the near-total absence of source-specific contaminant characterization as a prerequisite for agronomic use.

KEYWORDS

calcium and sulfur nutrition, cation exchange dynamics, phosphogypsum, saline-sodic soils, soil amelioration, soil physical properties

Introduction

Phosphorus (P) is an essential macronutrient for plant growth and development (1, 2). However, P deficiency in agricultural soils is the most common problem that must be addressed for boosting crop productivity and solving issues like nutrient deprivation, food security, and environmental sustainability challenges (2). On the other side, excessive or overuse of phosphate fertilizers not only may result in their accumulation in aquatic ecosystems, causing long-term eutrophication, but also has major environmental issues. Growing crop food products for a rapidly expanding population puts additional pressure on the crop ecosystems, which need additional P for boosting crop performance and productivity (3). Furthermore, P is also a non-renewable and limiting resource that must be used intelligently and efficiently to improve agronomic performance and favor the concept of environmental sustainability (3–5). Phosphogypsum (PG) is an important and underutilized by-product of phosphorus industries, which is highly enriched in calcium, phosphates, and sulfur, along with traces of fluorine, radionuclides (mainly ^{226}Ra , ^{210}Po , and ^{210}Pb), and heavy metals that possess significant environmental concerns (6, 7). Annual global PG generation currently exceeds 300 million tons, with cumulative stockpiles surpassing several billion tons worldwide (8, 9). Of this enormous volume, 58% is stockpiled on land, 28% is disposed of in coastal areas, and only 14% is reused or recycled (9). The United States Environmental Protection Agency (USEPA) has classified PG as a “Technologically Enhanced Naturally Occurring Radioactive Material” (TENORM), which directly reflects measurable radioactivity above natural background levels and has led to agricultural use restrictions in several jurisdictions. The environmental cost of unmanaged stockpiling is well-documented groundwater contamination, soil acidification, particulate emissions, and phosphorus leaching into surrounding ecosystems (10–15).

Despite the hazardous effects of PG as an environmental pollutant, several applications of PG have been well documented, including low-cost inorganic sorption materials (16), fly-ash PG binders, a modifier of bitumen for flexible road pavements (16), and alkali-activated fly ash paste (17). However, using PG in line with circular economy principles recommends their processing rather than stockpiling, and using PG as a substitute for natural gypsum has been intensively evaluated. Developing a secondary resource pathway would help manage resources effectively and support compliance with the Sustainable Development Goals (SDGs). With this objective, and following the principles of green chemistry, PG applications as soil conditioners or soil amendments provide a valuable resource in sustainable agriculture.

Currently, agricultural lands for farming practices are highly unstable due to rapid and unpredictable changes in climatic conditions. According to a report by the Food and Agriculture Organization of the United Nations (FAO), approximately 33% of the world's soil is moderately to highly degraded (18). Furthermore, excessive fertilizer applications have changed the physicochemical structure and functional profile of the soil. These include soil compaction (restricting root penetration and gaseous exchange),

progressive soil acidification from overuse of nitrogen fertilizers, soil organic matter depletion through accelerated mineralization, salinization and sodification from irrigation with poor-quality water, structural degradation manifested as clay dispersion and aggregate breakdown, and nutrient depletion from unbalanced fertilization (18). These degradation processes are reflected in measurable indicators, including increased bulk density, altered soil pH, elevated electrical conductivity, reduced aggregate stability, and declining microbial biomass, all of which adversely affect plant growth, development, and economic productivity. Traditional amendments such as lime and mined gypsum provide partial solutions but often fail to address the full spectrum of structural, chemical, and nutritional deficiencies in severely degraded soils, particularly in arid and semi-arid conditions (19). PG, with its high solubility, calcium and sulfate content, and relative abundance, has attracted growing interest as a soil amendment that could address multiple degradation constraints simultaneously while offering an agronomically beneficial outlet for an industrial waste stream (20, 21).

However, the path from promise to practice is not straightforward. One of the most critical concerns and also a major significant barrier in PG adoption for sustainable farming practices is the radionuclide content, particularly ^{226}Ra , ^{210}Po , and ^{210}Pb concentrated during phosphoric acid production. Furthermore, radon gas released from surface-applied PG may pose an inhalation risk to farm workers, whose intensity and severity are highly dependent on the production sources of PG and the application depth (6). A prolonged and frequent use of PG in farming practices may accumulate heavy metals like Cd, Pb, As, Cr, and Hg, and yet an accurate prediction of field-level data on their mobility, speciation, and bioavailability remains elusive.

The next barrier to widespread adoption of PG is methodological inconsistency or barriers, as several research papers have reported results in incompatible units (t ha^{-1} , g kg^{-1} , and % w/w), experimental durations ranging from weeks to single crop cycles, and contaminant characterization of the PG used is frequently absent, preventing the generalization of outcomes to a definitive conclusion. Furthermore, the transport costs add an economic dimension to this difficulty, since the $10\text{--}40 \text{ t/ha}^{-1}$ application rates required for effective soil conditioning render PG commercially unviable for smallholder farmers living far from production facilities (22).

The present review postulates that PG is an agronomically efficient, economically feasible, and environmentally acceptable circular-economy resource for restoring degraded soils and boosting crop productivity when applied at site-appropriate rates and in conjunction with complementary organic or biological amendments, as long as a thorough, source-specific contaminant risk assessment is carried out before use. The main objective of this review is to (i) develop the mechanisms by which PG enhances the physical, chemical, and nutritional properties of soil; (ii) critically evaluate crop productivity responses across various soil types and cropping systems; (iii) assess the risks to the environment and food safety associated with PG-derived radionuclides and heavy metals; and (iv) provide site-specific, evidence-based guidelines for safe and effective PG application within sustainable agro-ecological systems.

Literature methodology

Literature was searched across PubMed, Scopus, Web of Science, and Google Scholar using keyword combinations centered on PG and its interactions with soil properties, crop yield, heavy metals, radionuclides, soil microbiology, and circular economy themes during December 2025 to January 2026, and a narrative review was performed. The search covered publications from 2000 to 2025. The following primary keywords and their Boolean combinations were used: “phosphogypsum AND agriculture”, “phosphogypsum AND soil amendment”, “phosphogypsum AND soil salinity”, “phosphogypsum AND sodic soil reclamation,” “phosphogypsum AND crop yield,” “phosphogypsum AND heavy metals”, “phosphogypsum AND radionuclides,” “phosphogypsum AND soil microbiology,” and “phosphogypsum AND circular economy”. The inclusion criteria selected were peer-reviewed original research articles and review papers reporting quantitative data on PG effects on soil properties, plant nutrition, or crop yield. We excluded conference abstracts, research articles without peer review, and studies reporting PG effects exclusively in construction or non-agricultural industrial contexts. All studies meeting these inclusion criteria are incorporated in this review; no arbitrary numerical limit was applied to study selection. To map the intellectual structure and temporal evolution of PG research, a bibliometric analysis was performed using VOS viewer version 1.6.19 on publication data retrieved from PubMed using the keyword string “Phosphogypsum application in soil health management and crop quality and yield improvement”. The resulting network visualization (Figure 1A) identifies distinct thematic research clusters, including soil physical and chemical amelioration, saline and sodic soil reclamation, crop yield and plant nutrition, heavy metal immobilization and food safety, soil microbial community responses, and circular economy and industrial valorization. The overlay visualization (Figure 1B) shows a clear temporal shift in research focus, from physical and chemical soil effects dominant in earlier years toward food safety, circular economy integration, and environmental risk assessment in

more recent literature. This trajectory confirms that the present review addresses the most active and underserved frontier of current PG research.

Phosphogypsum: physicochemical mechanisms of soil structure improvement

PG potentially acts as a soil structure enhancer and significantly improves soil physical properties across diverse soil types and environments, addressing structural deformities and deficiencies in sodic, calcareous, arid, semi-arid, and degraded soils. The physicochemical basis of this improvement involves a cascade of reactions initiated by $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ dissolution. PG supplies soluble calcium that directly changes particle interactions and is central to how it alters soil structure. This calcium-driven flocculation promotes aggregation in salt-affected and weathered soils, and the magnitude of aggregate stability increases with PG dose. Ca^{2+} ions released into the soil solution act as polyvalent bridging cations between negatively charged clay mineral surfaces, compressing the electrical double layer around clay platelets and promoting face-to-face flocculation. Akanova et al. (23) reported that PG application enhances soil structure and physical properties by releasing Ca^{2+} ions, promoting flocculation and aggregate formation. It effectively addresses deficiencies in sodic soils by displacing Na^+ , improving macro-porosity, hydraulic conductivity, and bulk density through cation exchange (24, 25). Simultaneously, the increase in total electrolyte concentration (ionic strength) from SO_4^{2-} and Ca^{2+} ions further suppresses clay particle repulsion, accelerating aggregate formation. In sodic soils, Ca^{2+} displaces Na^+ from exchange sites through mass-action cation exchange, shifting exchangeable cation composition from dispersive (Na^+ -dominated) to flocculative (Ca^{2+} -dominated), with subsequent improvements in macro-porosity, hydraulic conductivity, and bulk density (Figure 2).

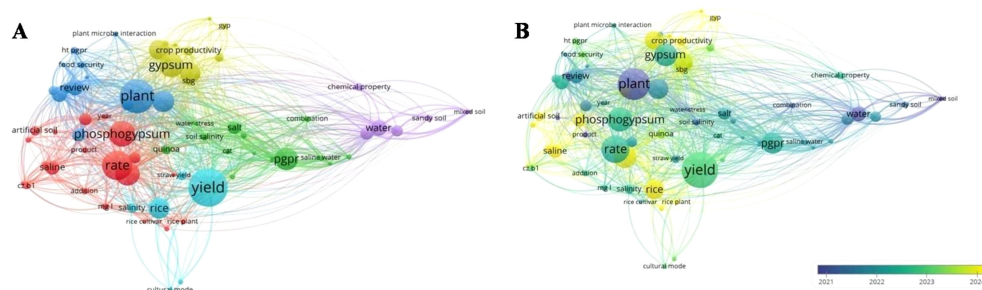


FIGURE 1

A bibliometric network map of scientific research on phosphogypsum application in soil health management and crop quality and yield improvement. The network visualization based on six colored clusters (A) indicates the relatedness of the scientific journals and an overlay visualization (B) specifies a period of the occurrence of the keyword from 2021 (blue) to 2024 (yellow). Six research clusters in panel A represent the following: (1) soil physical and chemical amelioration; (2) saline/sodic soil reclamation; (3) crop yield and plant nutrition; (4) heavy metal immobilization and food safety; (5) soil microbial community responses; and (6) circular economy and industrial valorization. Overlay (B) shows that research has shifted from physical/chemical effects (2021, blue) toward food safety, circular economy, and environmental risk assessment (2024, yellow), confirming that this review is aligned with current research priorities. This image is created from VOSviewer version 1.6.19, and collected data were obtained from PubMed database utilizing the keyword string “Phosphogypsum application in soil health management and crop quality and yield improvement”.

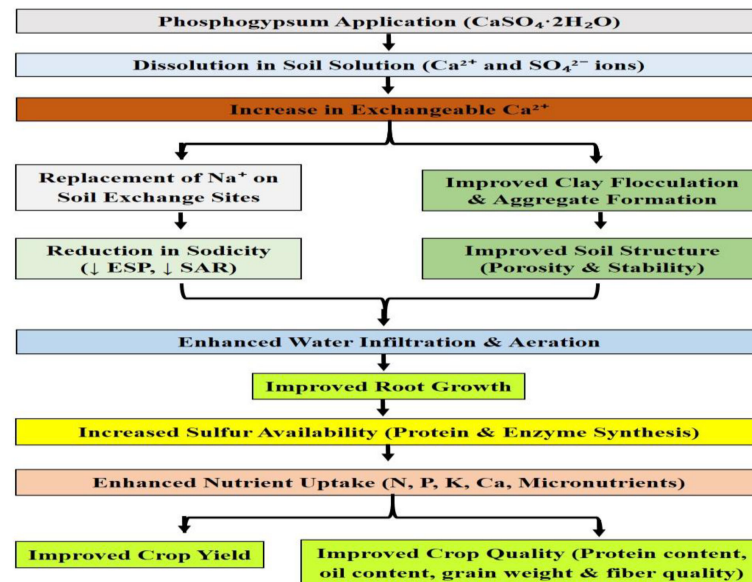


FIGURE 2

Mechanisms and agronomic benefits of phosphogypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$) application in soils: Schematic representation of the dissolution of phosphogypsum ($\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$), Ca^{2+} -mediated reduction of sodicity, and improvement in soil structure and water. $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ dissolution; Ca^{2+} - Na^+ and Ca^{2+} - Mg^{2+} exchange on soil colloids reducing ESP and EMP; Ca^{2+} -induced clay flocculation increasing aggregate stability, macroporosity, and hydraulic conductivity; reduced surface sealing; sulfate-S and Ca^{2+} supply to plant roots supporting protein synthesis; and subsoil Al^{3+} detoxification through Al-F complexation (AlF_2^+ species; 26) enhanced sulfur availability and nutrient uptake, resulting in improvements in root growth, crop yield, and crop quality.

Higher doses of PG (e.g., 45 t ha^{-1}) correlate with significant improvements in soil properties, such as a 5% reduction in bulk density and an 8% increase in total porosity (25) (Table 1).

While the benefits of PG are well-documented, some studies indicate that the effects may be temporary and dependent on specific soil conditions, suggesting that continuous monitoring and management are essential for sustained improvements. Makarova et al. (31) reported that structural changes in soil whose long-term irrigation was done with low-quality water could be effectively counteracted by PG application, as low-quality water caused soil structure deterioration through clay dispersion and migration, while PG restored structural integrity through enhanced soil aggregation and promoted loosening of soil compaction.

Several studies done so far have reported the key mechanism through which PG improves the soil structure and porosity (24, 25). The most common mechanism includes dissolution and ion release. As PG is rich in calcium sulfate dehydrate with residual phosphoric acid, upon dissolution, it releases Ca^{2+} and SO_4^{2-} . When compared to mined gypsum, PG on dissolution results in an acidic reaction medium that develops a higher electrolyte leaching solution, enhances the aggregation of the particles, and also improves soil hydraulic conductivity (24, 25).

Calcium-mediated flocculation and aggregate formation

Ca-based flocculation is another important mechanism where negatively charged soil clay particles form a stable aggregate with

positively charged cations, called flocculation, which encourages water infiltration and drainage, therefore preventing soil crusting. Flocculation is promoted by high levels of salinity (which may not be conducive to plant growth) and by the presence of cations that are strong flocculators. Adding polyvalent cations favors soil structure improvement, as divalent cations (like Ca^{2+}) can form cationic bridges between clay particles and organic carbon (36). The ability of dominant soil cations to flocculate soil clays is a function of their charge and size. Divalent cations (Mg^{2+} and Ca^{2+}) are much more powerful flocculators than monovalent cations (Na^+ and K^+). Several works done so far have reported the positive effects of PG on soil physical attributes such as aggregation, flocculation, and hydraulic conductivity in non-saline and non-sodic soils (36). These effects can extend into subsurface soil layers, resulting in greater growth and distribution of the plant root system.

Sodium displacement and soil sodicity mitigation

High pH, elevated concentrations of soluble salts, and the accumulation of Na^+ in saline and sodic soils collectively impose osmotic stress and Na^+ toxicity on crops while simultaneously reducing mineral availability through competitive ion interactions, most critically the displacement of K^+ by Na^+ at root absorption sites, which disrupts the cellular K^+/Na^+ ratio essential for normal metabolic function (37). Excess exchangeable Na^+ compounds cause these ionic and osmotic effects by driving aggregate slaking, clay swelling, and particle dispersion, which physically diminish

TABLE 1 Table summarizing the quantified effects of PG application on soil physical quality, including bulk density, aggregate stability, porosity, SAR, and ESP across 11 contrasting soil types and environments.

S. no.	Soil type/condition	PG rate (t ha ⁻¹)	Impact on soil physical quality	Reference
1	Saline-affected soil, Egypt (field, split-plot, 2 seasons, maize)	0, 3, 6, 9 t ha ⁻¹ + PGPR	PG 9 t ha ⁻¹ + <i>A. lipoferum</i> + <i>B. circulans</i> significantly reduced soil pH, EC, and ESP; improved urease and dehydrogenase enzyme activity; increased microbial communities and maize nutrient uptake over 2 growing seasons (2019–2020).	(27)
2	Salt-affected irrigated land, Egypt (wheat and rice)	~10.39 t ha ⁻¹ + vermicompost	Reduced ECE and base saturation (BS%); increased CEC and soil porosity. Wheat and rice grain yield significantly increased ($p < 0.05$) under combined PG + vermicompost.	(28)
3	Sandy loam soil, Egypt (wheat)	~25 t ha ⁻¹ PG + modified biochar	SOC increased +73.7%; soil available N + 130% vs. control; CEC significantly improved; available P + 102%; available K + 70%. Best results at PG + modified biochar (T7–T10 treatments).	(29)
4	Semi-arid hardsetting and crusting sandy soil, South Africa	3 t ha ⁻¹	MWD of macroaggregates: 2.23 mm (PG) vs. 1.36 mm (control). Significant improvement in aggregate size distribution and wet aggregate stability over two seasons under grass mulch + PG.	(30)
5	Saline ordinary chernozem, Ukraine (irrigated, long-term field)	3, 6 t ha ⁻¹	Bulk density reduced from 1.36–1.40 to 1.17–1.20 g cm ⁻³ ; soil porosity increased by up to 3.5%; enhanced water infiltration; reduced compaction in irrigated chernozem over multiple seasons.	(31)
6	Saline ordinary chernozem, Ukraine (irrigated and non-irrigated)	3, 6 t ha ⁻¹	SAR reduced by 69% under irrigation vs. only 10% without irrigation by year 3. Water permeability +0.66 mm min ⁻¹ (irrigated, 3 t ha ⁻¹). Toxic salts reduced to 0.41% (irrigated) and 0.38% (non-irrigated) by year 3. Salinity shifted to “slightly saline” category.	(32)
7	Poorly aerated film-mulched drip irrigation field, China	2–5 t ha ⁻¹	Bulk density reduced ~6.5%; soil water content +~11%; field capacity +~6%. Soil air permeability +142.3% under mulching conditions and +51.5% in inter-row zones. Verified from original paper figures.	(33)
8	Magnesium-affected irrigated soils, Central Asia (3-year trial)	3.3 and 8.0 t ha ⁻¹	Mitigated adverse structural effects of excess Mg ²⁺ across the top 60 cm; improved water infiltration and retention at both rates; increased crop water productivity and yield under Mg-affected sodic conditions.	(34)
9	Calcareous saline-sodic soil + low-quality irrigation water, Jordan	35 t ha ⁻¹	ESP reduced from 58% to below 10%; restored soil structure and reduced sodicity; improved hydraulic conductivity. PG + low-quality water combination effective for highly calcareous sodic soil reclamation.	(35)

SBD, soil bulk density; SP, soil porosity; MWD, mean weight diameter; ESP, exchangeable sodium percentage; SAR, sodium adsorption ratio.

PG, phosphogypsum; ESP, exchangeable sodium percentage; SOC, soil organic carbon; EC, electrical conductance; SBD, soil bulk density; SP, soil porosity; WP, water permeability; SWC, saturated water content; FC, field capacity; EMP, exchangeable magnesium percentage. PG generally lowers pH, which can increase nutrient solubility but may require monitoring. PG influences soil enzymatic activities critical to nutrient cycling. PG with organic amendments significantly increases soil NPK and SOC metrics.

Numeric values are reported directly from the original papers to allow direct comparison across studies and application rates.

aeration and hydraulic conductivity through blockage of conducting pores by dispersed clay particles, creating a soil environment where both root respiration and water infiltration are severely restricted (37). At an ESP (exchangeable sodium percentage) of approximately 30, application of sodic water to a clay soil reduced saturated hydraulic conductivity to between 1.2 and 3 mm h⁻¹, while in sandy loam soils, values declined to between 2.8 and 6.2 mm h⁻¹, with reductions driven primarily by clay dispersion and pore clogging rather than swelling alone (37). The combined effect of these chemical and physical constraints limits crop growth to the point where improvement of saline-sodic soil chemical and physical characteristics is a prerequisite, not merely a benefit, for restoring crop productivity potential (37).

PG directly targets these constraints through two integrated mechanisms. First, Ca²⁺ released from dissolving PG bridges negatively charged clay surfaces, reversing dispersion and promoting stable macro-aggregate formation through Ca²⁺-mediated flocculation (38). Second, Ca²⁺ displaces exchangeable Na⁺ from clay surfaces through direct cation exchange, reducing both ESP and SAR (sodium adsorption ratio) and mobilizing Na⁺ into the soil solution for removal under drainage and irrigation (38). PG

dissolves faster than mined gypsum, generates a leaching solution with higher electrolyte concentration, promotes more rapid particle aggregation, and recovers hydraulic conductivity more quickly at equivalent Ca²⁺ application rates—mechanistic advantages confirmed across multiple experimental systems (35).

Hasana et al. (38) evaluated five PG application rates from 0% to 200% of the gypsum requirement (GR) under incubation and leaching conditions, collecting soil and leachate samples at 7, 14, 21, 28, and 35 days, confirming a highly significant effect ($p < 0.001$) on soil pH, ESP, available P, and exchangeable Na⁺ and Ca²⁺. The Ca²⁺–Na⁺ exchange reaction was largely completed within 7–14 days following irrigation, with maximum Na⁺ and SAR removal occurring in the initial leachate volumes and declining progressively with subsequent watering. After the full 35-day leaching phase, the minimum SAR value of 4.14 mg·L⁻¹ was recorded at 200% GR, compared to 19.52 mg·L⁻¹ in the untreated control, confirming that simultaneous PG application and irrigation within this 7- to 14-day exchange window is critical for maximizing Ca²⁺–Na⁺ exchange efficiency. A comparative assessment of PG against CaCl₂ across seven application rates confirmed that both amendments reduced ESP by 90%; however, CaCl₂ removed 90% of total Na and soluble

salts while PG removed 79% and 60%, respectively, demonstrating that PG achieves equivalent sodicity correction but is less efficient in total salt removal, a practically important distinction where elevated salinity and sodicity co-occur in the same soil (35).

Aggregate stability and porosity

PG significantly influences aggregate stability and porosity, particularly when utilized in artificial aggregates and construction materials. The integration of PG into various mixtures enhances physical properties, mechanical strength, and durability, making it a valuable resource in sustainable construction practices. The use of PG in artificial aggregates, such as PG-based artificial aggregates (PMAs), has shown enhanced strength and stability. For instance, PMAs with added muck exhibited superior leaching stability and mechanical properties (39). Excess-sulfate phosphogypsum slag cement (ESPSC) combined with PG-based aggregates demonstrated better volume stability and mechanical properties compared to traditional cement mixtures (40).

Outbakat et al. (25) investigated the effect of PG on the physical properties of Luvisols and Cambisols collected from salt-affected soils in four regions in Morocco, including Chichaoua, Ras El Ain, Sidi Zouine, and Sed El Masjoune. The treatments consisted of different rates of PG (15, 30, and 45 t/ha⁻¹), natural gypsum (G) (15 t/ha⁻¹), and control. PG application presented a positive effect on other soil physical properties (soil hydraulic properties, total porosity, and bulk density), especially for the soils of Chichaoua and Ras El Ain regions. This work reported that PG application improved soil structure by promoting flocculent action provided by calcium. Outbakat et al. (25) demonstrated that PG application explained 93% of the water aggregate stability (WAS) variation ($R^2 = 0.93$), and this relationship was more pronounced in saline soils with a low level of salinity ($SAR < 6$). The increase in total porosity from PG was mostly due to an increase in total macropores, and PG promoted a greater concentration of calcium and sulfate compared to natural gypsum, having a favorable effect in increasing Ca^{2+} and reducing Na concentration in the soil.

Materechera et al. (30) evaluated the effect of PG application in restoring the soil aggregation, bulk density, and tensile strength in both amended and non-amended soils. PG application at 3 t ha⁻¹ greatly improved the stability of 2- to 4-mm macro aggregates under wetting conditions and increased the mean weight diameter (MWD) of soil aggregates (2.23 mm vs. 1.36 mm in control), suggesting higher structural resilience in semi-arid sandy soils (30). In semi-arid sandy loam soils that are prone to hard setting, PG-amended plots had lower aggregate tensile strength and bulk density than untreated soil, which increased porosity and decreased crust formation (30).

Water infiltration, hydraulic conductivity, and erosion control

PG plays a significant role in enhancing water infiltration, hydraulic conductivity, and erosion control in various soil types.

Its application, particularly when combined with other agents like polyacrylamide (PAM), has shown promising results in improving soil stability and reducing erosion. PG, when mixed with PAM, significantly increases the final infiltration rate, achieving improvements of three to five times compared to untreated soils (41). In sodic soils, the combination of PAM and PG maintains higher infiltration rates, even under conditions that typically lead to crusting and reduced water absorption (42). The integration of PG with microbial-induced carbonate precipitation (MICP) enhances the hydraulic conductivity of treated soils, allowing for better water movement and retention (43). Studies indicate that the permeability coefficient of PG can be reduced through microbial treatment, which strengthens particle connections and improves overall soil structure (43). The application of PG, especially in conjunction with PAM, has been effective in reducing soil erosion by two to four times compared to untreated controls (41). Enhanced erosion resistance is observed in treated slopes, with significant reductions in sediment loss during rainfall events (43).

Cochrane et al. (44) evaluated the synergistic application of PAM + PG in reducing soil erosion and surface run-off on Brazilian sandy soils. It was observed that surface application of PG and PAM or their synergistic treatments in combinations reduced soil erosion by an average of 90%, and runoff by 35% under simulated rainfall (25 mm h⁻¹), demonstrating that amendments are highly effective at preventing sediment loss. PG increased infiltration rates while reducing surface sealing by raising rainwater electrolyte content, which encouraged clay particle flocculation and stabilized surface aggregates. These findings highlight PG's ability to improve soil physical attributes in vulnerable sandy loam soils during high-intensity rainfall events (44). For erosion-prone subtropical sandy soils used in agriculture, all treatments performed similarly in terms of erosion control, indicating that even single-amendment applications (PG or PAM alone) offer significant conservation benefits (44).

Collectively, the evidence from simulated rainfall experiments confirms that the effectiveness of PG in maintaining infiltration and suppressing erosion is strongly modulated by soil texture, sodicity level, and whether PG is applied alone or in combination with PAM. Across four smectitic soils ranging in clay content from 10% to 62% and ESP from 2 to 20, both PAM mixed with PG and PG alone maintained final infiltration rates above 12 mm h⁻¹ and kept runoff and wash erosion at low levels compared to untreated controls; PAM mixed with PG consistently produced higher final infiltration rates and lower runoff than PG alone across all soil types, while with respect to erosion specifically, PAM + PG was more effective than PG alone in loamy sand and clay soils but produced comparable sediment loss reduction in loam soils (42).

Field-scale structural improvements under irrigation stress

At the field scale, dose-dependent SAR reduction by PG is well documented across contrasting soil environments. On irrigated chernozem soils receiving mineralized irrigation water at 2.3–3.0

g L^{-1} , PG applied at the absorption-norm dose reduced SAR by 74.5% compared to the irrigated control, while the dose calculated by the coagulation limit without irrigation reduced SAR by 23%, with the lowest absolute SAR values recorded in the highest-rate irrigated plots (32). Across four Moroccan salt-affected soils (Luvisols and Cambisols), 45 t ha^{-1} of PG produced an 8% increase in total porosity in Ras El Ain soil, and bulk density was 5% lower at the same rate in Chichaoua soil compared to the control, with water aggregate stability strongly correlated with PG dose ($R^2 = 93.41\%$, 25). In terms of salt displacement at the field scale, doses of 30 and 45 t ha^{-1} were most efficient in removing leachable salts from both saline and saline-sodic soils, with trace element concentrations in leachate remaining below recommended limits for drinking and irrigation water under the prevailing alkaline conditions (25). Structural improvement attributable to PG-mediated aggregate stabilization in chernozem soils subject to secondary salinization from mineralized irrigation includes recovery of water permeability from 1.10 to 2.60 mm min^{-1} and reduction in bulk density to 1.17–1.20 g cm^{-3} in non-irrigated variants after re-application at 3–6 t ha^{-1} (31), alongside a Na/Ca exchange ratio reduced to 1.51 under irrigated spring application at 3 t ha^{-1} and exchangeable sodium decreased to 3.7% of total cation exchange capacity with irrigation. Water percolation associated with leaching raised saturation index values in drainage water, signaling potential for mineral precipitation in shallow groundwater, a secondary risk requiring site-specific monitoring under intensive PG-assisted leaching programmes (25).

Beyond SAR reduction, Onopriienko et al. (32) reported that PG application on ordinary chernozem soils of Ukraine's northern steppe where baseline toxic salt content ranged from 0.35% to 0.48% and exchangeable sodium stood at 3.64% with an unsatisfactory physical condition before treatment produced measurable improvements in salt regime and physical condition by the third year after application, with the magnitude of improvement determined by irrigation regime and application rate. Multi-year results from the same experimental series further show that SAR reduction reaches 74.5% under irrigation and 23% without irrigation across all study years, confirming the superior reclamation efficiency of irrigated PG application at the absorption-norm dose (32). Beyond sodicity correction and physical improvement, PG stabilizes soil aggregates through Ca^{2+} – Na^+ exchange while replenishing plant-available sulfur via residual sulfate ions; sulfate leaching to subsoil layers further enhances root-zone calcium supply and has been shown to reduce subsoil aluminum toxicity in tropical no-till systems (32, 45).

The consequences of high- Mg^{2+} irrigation water on soil structural function were rigorously documented by Vyshpolsky et al. (46) in a 4-year field experiment conducted in Southern Kazakhstan, where canal irrigation water carried a $\text{Mg}^{2+}:\text{Ca}^{2+}$ ratio of 1.30–1.66, conditions representative of more than 30% of irrigated lands in the region characterized by low infiltration rates and declining hydraulic conductivities. Single-dose PG application at 4.5 t ha^{-1} produced an 18% decrease in exchangeable magnesium percentage (EMP) in the surface 0–20 cm soil after the first crop harvest, while the higher rate of 8 t ha^{-1} reduced EMP by 31%, confirming a clear dose–response relationship in Ca^{2+} -mediated Mg^{2+} displacement. The structural recovery enabled by this ionic correction translated directly into agronomic output: 4-year average

cotton yields reached 2.6 t ha^{-1} under 8 t ha^{-1} PG and 2.4 t ha^{-1} under 4.5 t ha^{-1} PG, compared to only 1.4 t ha^{-1} in the untreated control, with economic returns from PG treatments nearly double those of the control. Importantly, Vyshpolsky et al. (46) noted that exchangeable Mg^{2+} levels began to recover 4 years after a single application, particularly at the lower rate, establishing that sustained structural improvement under high- Mg^{2+} irrigation requires repeated PG application at 4- to 5-year intervals to maintain optimal ionic balance. Onopriienko et al. (47) investigated the agro-ecological status of soils exposed to more than 50 years of continuous irrigation and applied PG at doses determined by three agronomic criteria, including the amount of exchangeable Na^+ requiring displacement, the coagulation limit, and the sodium absorption norm corresponding to incoming irrigation water and provided parallel evidence under sodic irrigation conditions. Positive changes in the anion–cation balance were confirmed in all PG-amended treatments through multi-year monitoring of the composition of water-soluble salts. Beyond ionic correction and structural recovery, PG application under irrigation stress also modulates crop nutrient dynamics in ways that reflect altered soil physicochemical conditions. In a completely randomized experiment, Quintero et al. (48) applied PG at rates of 0, 2, 6, and 20 t ha^{-1} in factorial combination with nitric acid acidification to an initially sodic soil under drip-irrigated commercial tomatoes. The highest PG rate increased dry matter yield, attributable in part to improved water use efficiency arising from better soil structural conditions. Although PG decreased K, Mg, and P concentrations in shoots and P and Cu in fruits through ionic antagonisms, these negative interactions did not result in yield reductions even at a dose simulating 20–30 years of cumulative application, confirming the structural and nutritional resilience of PG-amended sodic soils under drip irrigation. Quintero et al. (48) also found that the concentration of all micronutrients in shoots and Fe, Cu, and B in fruits increased when nitric acid was applied in tandem with PG treatments, simulating acidification cleaning methods used in drip irrigation systems. This suggests a synergistic improvement in micronutrient nutrition.

The environmental dimension of PG application under irrigation was addressed by Li et al. (49), who investigated fluorine dynamics in southern chernozems across non-irrigated, irrigated, and irrigated-reclaimed (PG-amended) plots within irrigation areas of the Odessa region. Total bulk fluorine content was highest in non-irrigated chernozems but did not exceed maximum permissible concentrations, while irrigation reduced bulk fluorine in topsoil through dissolution and downward migration within the soil profile. Li et al. (49) reported that the combined application of phosphorus fertilizers with manure reduced fluorine content in soils, lysimetric waters, and crop tissues, and recommended that when irrigation of chernozem soils is resumed after a break, increased doses of mineral fertilizers including phosphorus, should accompany PG application to maintain fluorine within safe agronomic and ecological bounds.

PG application at a rate of 1.4–6 t ha^{-1} demonstrated the mitigative effects on chernozem soils in Northern Steppe of Ukraine that were degraded due to irrigation waters. In non-irrigated soils, PG application at 3–6 t ha^{-1} resulted in significant

improvement in soil structure parameters with reduced bulk density (1.40 to 1.17–1.20 g cm⁻³), improved soil permeability (from 1.10 to 2.60 mm min⁻¹) and enhanced soil porosity (by up to 3.5%), which overall resulted in improved soil structure along with better electrical conductivity in saline chernozems (31). However, soil particle composition was reported to be more resistant toward PG application than to irrigation impacts. Irrigation with Class II water [defined under Ukrainian irrigation water quality standards as water with mineralization of 1.0–1.5 g L⁻¹ and electrical conductivity of 0.75–1.25 dS m⁻¹; applied at 1,500 m³ ha⁻¹ in this study; (31)] reduced physical clay content by 0.06%–0.12%, while combined application of irrigated water with PG resulted in minor redistributions between physical sand and clay fractions.

Combined PG and organic amendment approaches

Co-application of PG with organic amendments commonly and substantially increases available nitrogen, phosphorus, and potassium relative to control treatments, with reported effects ranging from large percent increases to multi-fold enrichments (50, 51). These increases are observed across pot, greenhouse, and field studies and are influenced by amendment type and rates. However, the size and consistency of SOM and NPK responses depend on amendment type, application rates, soil type, and time, and combined treatments often translate into higher crop yields. The synergistic application of biochar (BC) with PG has promising effects, PG + BC synergistic treatment complements the function of each other where alkalinity contributed by BC is used to counteract the acidity of PG in soil, and mineral nutrient contributed by PG enhances and supplements the BC benefits in promoting soil fertility, nutrient dynamics, and ecological sustainability through providing a novel platform for PG valorization, waste management, and a cost-effective solution for nutritional security (33). Differences among studies reflect experimental scale (pot versus multi-year field), amendment composition (BC versus fresh manure), and PG and organic matter rates. BC + PG produced very large SOC and nutrient gains in loamy sand, while fresh manure and turkey/poultry litter produced steady SOM and multi-fold nutrient increases in heavier soils (29, 52, 53). Mahmoud et al. (54) reported an experimental design with PG supplementation with poultry manure (PM) in calcareous soil used either alone or in combination with different treatment rates such as PG₁ (4.20 g kg⁻¹), PG₂ (6.30 g kg⁻¹), PM₁ (4.20 g kg⁻¹), and PM₂ (6.30 g kg⁻¹), plus combinations and control for improving barley crop yield and performance. This work demonstrated the PG + PM synergistic effect on restoring or boosting microbial performances or changing the microbial composition with improved microbial biomass carbon (MBC), organic matter content, and soil NPK content and availability. Multi-year experiments show larger humus accumulation when PG is combined with substantial organic matter inputs (for example, 10 t ha⁻¹ PG + 40 t ha⁻¹ OM gave the greatest humus accumulation after 4 years). However, the increase in nutrient content is highly dependent on amendment type and their application rates (52). A novel

biomeliorant made of PG, bovine manure, and camel thorn dramatically enhanced the characteristics of saline and damaged soils, based on field tests conducted in southern Kazakhstan. When this mixture was applied, the humus content increased by 59%–82%, soil aggregation improved, and the pH of the soil decreased to almost neutral levels. Under arid zone conditions, these modifications increased the displacement of sodium ions from the soil complex and improved water permeability, leading to yield improvements (55).

Interestingly, Li et al. (56) evaluated the dose-dependent modulation of PG in soil microbial activities measured in terms of soil respiration, enzyme activity changes, and microbial community impact. The experimental design developed was an indoor incubation experiment testing five PG application rates, namely, 0%, 0.01%, 0.1%, 1%, and 10%. This work demonstrated the direct dose-dependent effect of PG supplementation on soil properties with higher PG doses resulting in progressive changes in soil microbial community composition and environment with minimum effect at lower doses (0.01%) in physio-chemical and biological properties and best observed with (10% PG). However, at higher doses of PG, microbial respiration rate decreases, showcasing reduced microbial metabolic activity. This becomes clearer as 0% PG treatment results in decreased catalase, urease, and phosphatase activities, but increased sucrase activity, while 0.01% or 0.1% PG treatment causes improved urease activity.

Collectively, the evidence on PG and soil microbial communities converges on two conclusive positions. First, the consensus is dose-defined: PG at ≤0.1% soil weight preserves or improves microbial enzyme activity, whereas PG at 10% significantly reduces the abundance, diversity, and evenness of both soil bacteria and fungi, increases water-soluble fluoride 19.84-fold, and suppresses catalase, urease, and phosphatase activities (56). Second, the apparently contradictory finding that PG harms microbial communities in some studies but improves them in others is explained by starting soil condition and application rate: in acid soils, high PG doses compound existing chemical stress by increasing fluoride and ionic load, suppressing microbial activity; in alkaline sodic soils, agronomic PG rates correct the root cause of microbial stress by displacing Na⁺ with Ca²⁺ and shifting pH toward neutral, establishing conditions in which microbial communities recover.

Magnesium-affected soil reclamation

On secondary-salinized chernozem soils of Ukraine's northern steppe irrigated with mineralized water from the Samara River for over 50 years, PG demonstrated complementary benefits in correcting sodium-driven sodicity. Research confirmed that irrigation with mineralized water increased exchangeable sodium in the soil exchange complex to 5% and in the aqueous soil extract by 30%. PG applied at rates of 1.4 and 3 t ha⁻¹ in spring under irrigation reduced the Na/Ca ratio from a 3-year average to 2.53 and 1.51, respectively, while autumn application at 6 t ha⁻¹ without irrigation decreased exchangeable sodium to 2.3% of total exchangeable cations, compared to 3.7% under spring irrigated application. According to the

anion composition, soils in PG-treated plots shifted from soda-sulfate chemistry (control, no PG, and no irrigation) to sulfate-type salinity chemistry, while a gradual reduction in sulfates was observed by 1.5% in the second year and 7.5% in the third year after PG application (32). Based on SAR, PG-treated plots were classified as slightly saline, while control plots without PG retained moderate salinity, confirming that PG's reclamation benefit on long-term irrigated chernozem soils extends beyond structural improvement to fundamental correction of the ionic regime responsible for secondary salinization (32).

PG's reclamation potential extends to highly degraded soils in arid oasis environments. Gabisi et al. (57) evaluated PG at 12.5, 25, and 50 t ha⁻¹ combined with cattle manure at 37.5 t ha⁻¹ on degraded Tunisian oasis soils and found that all PG rates improved germination of turnip, jute mallow, and oat compared to the manure-only control, with the highest dose (50 t ha⁻¹) producing the greatest germination rate across all three crops. Porosity increased progressively with PG rate from 52% in the control to 59%, 61%, and 63% at increasing PG doses, improving seedbed conditions and water retention proportionally to application rate. Cadmium levels in all crops remained below the 7 parts per million (ppm) toxicity threshold, and Ra-226 radioactivity in the applied Tunisian PG measured 214.45 Bq/kg below the USEPA threshold of 370 Bq/kg (10 pCi/g) for agricultural use, although this limit applies specifically to this Tunisian PG source and cannot be extrapolated to PG from other phosphate rock origins, which vary substantially in radionuclide content depending on the uranium concentration of the parent ore (6, 10).

Phosphogypsum as a source of macronutrients and micronutrients

PG supplies essential macronutrients, primarily calcium (Ca), sulfur (S), and phosphorus (P), while simultaneously improving soil quality through regulation of acidity, modification of aluminum forms and amorphous ferric oxide content, reduction of heavy metal leaching, and minimizing soil erosion (37, 45). These nutrient supply functions are soil type dependent and have been demonstrated across a wide range of agronomic systems. PG treatment of Jeju volcanic ash soils produced the highest exchangeable Ca and S contents alongside a marked decrease in exchangeable aluminum and Al saturation; in Andisols with high Al and Fe, PG increased adsorbed S and enhanced long-term sulfur retention, while in the non-Andisol Gangjeong soil, sulfur from PG was retained predominantly in exchangeable forms, confirming PG as a readily plant-available sulfur source whose effectiveness varies with soil mineralogy. Similarly, Bouray et al. (58) reported that PG at 9 t ha⁻¹ combined with lime produced maximum lucerne yields and P and S uptakes in two contrasting soils, with exchangeable Al decreasing in both soils at 1 t ha⁻¹ of PG compared to the control (Table 2).

The dual role of PG in supplying Ca and S while reducing subsoil constraints has been further confirmed in long-term no-till tropical systems, where surface application of PG, particularly when combined with lime, raised exchangeable Ca²⁺, K, Mg, N, and S

concentrations in all soil layers down to 0.40 m, demonstrating the downward mobility of Ca²⁺ and SO₄²⁻ from PG and its capacity to address subsurface acidity and nutrient deficiency that typically restrict root development and water and nutrient access in these systems (60).

Across legume and vegetable crops, PG has consistently raised plant-available calcium in both soil and tissues. Surface application of PG at 2.1 t ha⁻¹ enhanced Ca content in common bean leaves from 16.3 to 20.0 g kg⁻¹, demonstrating its effectiveness as a Ca source in acidic tropical soils (65). In tomato, PG application raised Ca²⁺ levels in both shoots and fruits, with fruit calcium concentration increasing from 10.29 to 25.66 mg kg⁻¹ DW at 10% amendment rate, thereby enhancing fruit integrity and reducing susceptibility to blossom-end rot, a calcium deficiency disorder responsible for yield losses of up to 50% in severe cases (66). In Andosols, PG boosted water-soluble calcium from 1.7 to 5.2 cmol_c L⁻¹ in nursery media without altering soil pH, with the effect most pronounced in soils with reduced allophane content where Ca adsorption was limited; notably, PG outperformed lime in delivering soluble calcium, making it a practical alternative for seedling establishment in acidic volcanic soils (67).

In cereal and forage systems, PG has demonstrated a capacity to increase calcium availability not only in surface layers but also at depth. Nascimento et al. (68) reported that soil exchangeable Ca²⁺ increased linearly in both the 0–0.2 m and 0.21–0.4 m layers with PG application up to 1.2 t ha⁻¹, with alfalfa achieving maximum leaf Ca concentration at 9.5 t ha⁻¹, confirming its value as a calcium source for deep-rooted legumes. Similarly, Vicensi et al. (64) documented that PG raised leaf Ca content in corn, wheat, and soybean across all treatments, with the strongest response in corn leaf Ca rising from 4.53 to 5.26 g kg⁻¹ as PG rate increased from 0 to 1.2 t ha⁻¹. Soratto and Crusciol (65) further confirmed that 2.1 t ha⁻¹ of PG raised Ca concentrations in leaves of soybean, rice, common beans, and maize simultaneously, with maize leaf Ca increasing from 2.3 to 3.1 g kg⁻¹, reflecting improved cell wall integrity and nutrient translocation across multiple crops in a single experiment (Table 3).

Sulfur supply mechanism and soil profile penetration

Because PG is composed primarily of calcium sulfate dihydrate, it delivers sulfur to soil exclusively as sulfate (SO₄²⁻), the form directly absorbed by plants. Unlike organic sulfur, dominant in surface soil layers, sulfate from PG moves readily through the soil profile under water percolation, making PG an effective sulfur source for subsurface layers where sulfur deficiency frequently limits crop production (60). Vicensi et al. (64) confirmed that sulfur concentrations increased linearly with PG application rates to a depth of 0.8 m under no-till, with PG raising leaf S concentrations in corn, wheat, and soybean across all treatments, corn achieving the maximum leaf S content of 5.56 g kg⁻¹ at 12 t ha⁻¹ PG. Long-term combined application of lime and PG over 16 years in no-till tropical soils not only raised SO₄²⁻-S throughout the profile

TABLE 2 Table documenting changes in soil agrochemical properties—including pH, organic carbon, CEC, and macro- and micronutrient availability following PG application alone or in combination with lime, biochar, organic manure, and microbial inoculants.

S. no.	PG amount used	Soil agrochemical properties improved	Reference
1	1.5, 3, 4.5 t ha ⁻¹ field, barley, Morocco, 2 years	Decreased soil pH; increased soil P and S content up to +42% at 4.5 t ha ⁻¹ . Grain yield ↑ 21% (1.5), ↑ 34% (3), ↑ 39% (4.5 t ha ⁻¹). No effect on heavy metal content in grain or shoots. PG more efficient than natural gypsum at equivalent rate.	(51)
2	2.5 t ha ⁻¹ PG + 2.5 kg m ⁻² modified biochar sandy loam, Egypt, wheat	SOC +99%; available N + 161%; available P + 102%; available K + 70%; CEC significantly increased vs. NPK-only control. Best: PG + modified biochar combination (T7–T10).	(29)
3	5.8 t ha ⁻¹ PG Ultisol column leaching, China (short-term)	Masud et al. (59) compared alkaline slag (AS) vs. PG. PG alone did not raise pH from 4.96 to 5.11; that effect was from alkaline slag. PG mainly facilitated SO ₄ ²⁻ mobility and Ca ²⁺ leaching to subsoil. AS was more effective than PG at increasing pH.	(59)
4	PG with lime and silicate Rhodudult, no-till upland rice, Brazil	Improved subsurface acidity; reduced Al ³⁺ toxicity through high Ca ²⁺ and SO ₄ ²⁻ mobility to subsoil. Enabled better root development in upland rice on tropical Oxisol. Surface application of lime + silicate + PG enhanced both topsoil and subsoil conditions.	(60)
5	PG + <i>Bacillus velezensis</i> Pb-contaminated soil, China (<i>Cynodon dactylon</i>)	Increased soil pH from 5.61 to 5.77; reduced Pb mobility and bioavailability; improved rhizosphere soil nutrients and microbial community composition under Pb contamination. Combined PG + <i>B. velezensis</i> more effective than PG alone.	(61)
6	PG 2.1 t ha ⁻¹ + lime no-till soybean–oat–sorghum rotation, Oxisol, Brazil (15 years)	Enhanced Ca ²⁺ mobility to subsoil without surface pH alteration; reduced Al toxicity in subsoil over 15 years; consistent soil chemical improvement and crop yield improvement across all rotation crops.	(62)
7	PG 4.20 t ha ⁻¹ + poultry manure calcareous soil, pot experiment, barley, Egypt	Reduced soil pH from 8.08 to 7.64 in calcareous soil; increased available N, P, K, and OM; significantly improved MBC; enhanced barley yield and soil fauna diversity (Shannon index +19%–28%).	(54)
8	PG 1–5 t ha ⁻¹ + cow manure + PGPR saline calcareous greenhouse, maize, Saudi Arabia	Reduced soil pH; ECe decreased with inoculation; available P increased significantly; N, P, K, Fe, Mn, Zn, and Cu availability improved; Cd and Pb bioavailability to maize reduced. Best: PG 5 t ha ⁻¹ + co-inoculation.	(63)
9	PG 0, 1, 3, 9 t ha ⁻¹ acid soils, glasshouse, lucerne, New Zealand	P and S bioavailability increased with PG rate. Exchangeable Al decreased at 1 t ha ⁻¹ . Olsen P increased +8 mg kg ⁻¹ (PG) and +6 mg kg ⁻¹ (PS control). Maximum yield and P/S uptake under PG 9 t ha ⁻¹ + lime.	(58)
10	PG 3–12 t ha ⁻¹ - Typic Hapludox, no-till, Brazil (corn–wheat–soybean)	S-SO ₄ ²⁻ increased to 0.8 m depth; Ca ²⁺ increased; Mg ²⁺ migrated downward at all rates. Standard dose (6.38 Mg ha ⁻¹) = best balance. Double dose (7.42 Mg ha ⁻¹) = greater residual sulfate but reduced Mg ²⁺ and K ⁺ .	(64)

SOC, soil organic carbon; MBC, microbial biomass carbon; CEC, cation exchange capacity.

Studies span acid, calcareous, saline, and sodic soils across tropical, subtropical, Mediterranean, and arid environments.

but also reduced ¹⁵N loss and improved ¹⁵N recovery in crops by regulating soil nutrient availability and nitrogen cycling genes, demonstrating that PG's sulfur supply has indirect positive effects on nitrogen use efficiency beyond its direct role as a sulfate source. During the active growth phases of cereals, the sulfate supplied by PG promotes sulfur-dependent metabolic processes, including protein synthesis and chlorophyll development, with these effects particularly pronounced when PG is used in crop rotations that alternate between cereals and legumes, as the residual sulfate from a previous PG application continues to benefit the subsequent crop (23). Al-Enazy et al. (63) reported that PG paired with cow manure and microbial inoculants in calcareous soil improved macronutrient uptake N, P, and K alongside availability of micronutrients, including iron, manganese, zinc, and copper in maize, with bacterial co-inoculation additionally reducing cadmium and lead bioavailability to plants. Sulfur response to PG is rate- and crop-dependent. Vicensi et al. (64) found that while corn yield showed a quadratic response to PG with maximum technical efficiency at 6.38 t ha⁻¹, soybean yield was unaffected by PG rate, and Mg²⁺ migrated downward through the soil profile regardless of application rate or splitting, a consistent limitation of PG use that must be factored into fertilization planning in Mg-deficient soils. A double dose of PG (7.42 t ha⁻¹) had a greater residual sulfate effect in subsurface

layers but caused decreases in Mg²⁺ and K⁺ levels, confirming that the standard recommended PG dose provides the best balance between sulfur supply and cation retention, and that excessive PG application in low-acid soils risks nutrient imbalance without proportional yield benefit (72).

Sulfate-phosphate competition mechanism

PG addresses this constraint through a well-documented competitive displacement mechanism: the SO₄²⁻ released from PG competes with phosphate ions for adsorption sites on Fe and Al oxide surfaces, reducing P fixation and increasing the concentration of plant-available phosphate in soil solution (58). This mechanism is particularly effective in P-fixing soils with high amorphous Fe and Al oxide content, where sulfate competes directly for the same ligand-exchange sites, releasing previously fixed phosphate back into the soil solution (64). Bouray et al. (58) further confirmed in a 60-day incubation study using two acid soils from New Zealand that PG addition at rates of 1–9 t ha⁻¹ combined with phosphate rock significantly increased Olsen P and Ca saturation while reducing

TABLE 3 Table compiling the specific effects of PG application on soil macro- and microelement dynamics, microbial biomass carbon, enzyme activity, and heavy metal behavior across diverse soil and crop systems.

S. no.	PG amount used	Impact on soil macro- and microelements (verified from original paper)	Reference
1	1.5, 3, 4.5 t ha ⁻¹ field, barley, Morocco, 2 years	Enhanced barley shoot and grain uptake of N, P, K, Ca, Mg, and S at all rates. PG sources (PG1 and PG2) increased grain yield by 61% and 59% vs. control across rates. S uptake in shoots significantly increased. No significant effect on heavy metal (Cd, Cr, Pb, and Zn) content in shoots or grain.	(51)
2	4.20 g kg ⁻¹ + poultry manure calcareous soil, pot, barley, Egypt	Combined PG + poultry manure increased available N, P, K, and OM in calcareous soil. MBC significantly improved. Soil fauna diversity (Shannon index) +19%–28% vs. control. Best: PG + PM at highest combined dose.	(54)
3	Modified PG artificial soil (HLRT vs. ILRT) China, buckwheat	Bulk density reduced 3.8%–6.9% under HLRT; moisture +22%; phosphorus mineralization (S-NP) 7.53 U/g under HLRT. Buckwheat yield +130.6% vs. control. Heavy metal BCF in grain <0.3. Pb exceeded national standard due to background soil, not PG.	(19)
4	PG 5 t ha ⁻¹ + <i>Bacillus velezensis</i> Pb-contaminated soil, China (<i>Cymodon dactylon</i>)	Improved microbial-mediated nutrient cycling; urease and phosphatase activities improved. Rhizosphere soil nutrients improved. Pb bioavailability reduced; soil pH 5.61→5.77. Combined PG + <i>B. velezensis</i> reduced Pb mobility significantly.	(61)
5	PG 1–5 t ha ⁻¹ + cow manure + microbial inoculants calcareous greenhouse, maize	N, P, K, Fe, Mn, Zn, and Cu uptake improvement in maize. Available P increased significantly with PG + inoculation. Cd and Pb bioavailability to maize reduced. Soil pH reduced; ECe decreased with inoculation.	(69)
6	Phosphogypsum calcareous field soil, Jordan (tomato and green pepper)	Elevated total concentrations of Cd, Cr, Pb, and Zn in tomato and green pepper tissue. Health risk assessment indicated potential risk at high PG rates due to radionuclide and heavy metal content in phosphate-derived PG.	(70)
7	PG 10 t ha ⁻¹ saline-sodic soils, NE China (rice)	Increased soluble Ca ²⁺ in 0–20 cm topsoil. Structural improvement of sodic soil via Ca ²⁺ –Na ⁺ exchange. Improved rice growth and grain yield; SAR and ESP reduced progressively with PG.	(71)
8	PG + dolomite surface application Typic Hapludox, no-till, Brazil	Enhanced subsoil Ca ²⁺ and S-SO ₄ ²⁻ availability; improved Ca: Mg balance at depth; increased corn, wheat, and soybean nutrition in subsurface layers vs. dolomite-only control.	(65)
9	0, 0.01%, 0.1%, 1%, 10% PG acid soil incubation, China (7 and 28 days)	New row: Low PG (0.01%–0.1%) improved urease; 10% PG suppressed catalase, urease, phosphatase; sucrase increased at 10%; CO ₂ respiration declined with rate. Fluoride and S bioavailability increased with rate. Non-linear enzyme dose response confirmed.	(56)
10	PG 9 t ha ⁻¹ + PGPR saline soil, Egypt (maize, field, 2 seasons)	New row: PG + <i>A. lipoferum</i> + <i>B. circulans</i> significantly increased N, P, K, and micronutrient uptake in maize. Soil enzyme activity (urease, dehydrogenase) improved. Soil pH, EC, and ESP decreased. Chlorophyll and carotenoid content increased. Combined PG + PGPR superior to either alone.	(27)

BCF, bioconcentration factor; HLRT, hierarchical land reclamation technique.

PG supplies Ca, S, and P directly and enhances availability and uptake of N, P, K, and Mg. Combining PG with organic amendments can increase micronutrient availability by altering soil chemistry and microbial activity. Improvements often arise via enhanced cation exchange capacity, pH modulation, and enzymatic nutrient cycling.

The table highlights both the nutritional benefits of PG-derived Ca, S, N, P, K, and micronutrients and the potential risks of heavy metal accumulation at excessive application rates.

exchangeable Al, confirming that PG enhances phosphate rock dissolution and P availability simultaneously in acid soils.

Phosphorus availability in land reclamation and modified PG systems

In modified PG artificial soil (MPG-soil) prepared for land reclamation under the hierarchical land reclamation technique (HLRT), PG amendment significantly increased available phosphorus, with phosphorus mineralization reaching 7.53 U/g S-NP activity, the highest recorded across all treatments; however, organic matter content was simultaneously reduced, indicating that PG-based land reclamation systems require supplemental organic fertilization to sustain long-term phosphorus cycling and soil biological function (19). While PG improves P availability at agronomic rates, excessive or poorly managed applications carry environmental risks. Fluorine released from PG exists mainly as residual CaF₂ in the soil matrix and poses minimal ecological risk at moderate application rates; however, the co-presence of fluoride

and phosphate in PG-amended soils requires monitoring because F⁻ competes with phosphate for soil adsorption sites and can alter P dynamics unpredictably at high PG doses. Additionally, the residual P₂O₅ fraction in PG can leach into groundwater under high rainfall or irrigation, making application rate management critical in porous or sandy soils (19, 25).

Influence on secondary and trace elements

Beyond the primary macronutrients calcium, sulfur, and phosphorus, PG application introduces secondary and trace elements into the soil system whose effects range from beneficial nutrient supplementation to potential contamination of the food chain. PG contains residual quantities of fluoride (F⁻), trace metals including cadmium (Cd), lead (Pb), chromium (Cr), nickel (Ni), and copper (Cu), as well as naturally occurring radionuclides, all of which are inherited from the parent phosphate rock during wet-process phosphoric acid production. At the same time, PG serves as a

secondary source of micronutrients, including zinc, iron, manganese, and copper; Nayak et al. (73) reported that DTPA-extractable Zn, Fe, Mn, and Cu were higher in PG-amended soils due to the acidifying effect of PG in the rhizosphere, which increased micronutrient solubility and plant availability (Table 3). Bouray et al. (58) further confirmed using visual Minteq speciation modeling across three New Zealand acid soils that Al^{3+} and hydroxylated Al fractions decreased significantly at 1 and 3 t ha⁻¹ of PG compared to control, and that total dry matter yield of lucerne was better explained by Al^{3+} , Al-F, and Al-DOM fractions than by exchangeable Al alone, confirming that PG reduces phytotoxic Al species rather than total Al.

Beyond its role in Al detoxification, PG can function as an *in situ* immobilization agent for heavy metals in contaminated agricultural soils. Mahmoud and Abd El-Kader (74) demonstrated that PG applied alone at 10 and 20 g dry weight kg⁻¹ soil significantly reduced Pb, Cd, and Zn uptake in canola plants grown in contaminated soil by forming surface metal complexes at Ca-OH, Fe-OH, Al-OH, and Si-OH sites on the PG surface; PG alone achieved the maximum metal immobilization efficiency, while the combined PG + rice straw compost (1:1 ratio) at 10 g kg⁻¹ increased canola dry biomass by 66.8% demonstrating that PG is the preferred single amendment for metal stabilization, while the PG-compost combination is optimal where both remediation and crop production are required simultaneously. PG also has a documented capacity to remove heavy metals from aqueous solution through pH-dependent adsorption, a property with relevance to both wastewater treatment and *in situ* soil remediation. Es-Said et al. (75) characterized the simultaneous adsorption of Cd^{2+} , Cu^{2+} , and Zn^{2+} by PG from aqueous solution using response surface methodology and confirmed that adsorption efficiency is pH-dependent, with maximum removal occurring in alkaline conditions, and that PG exhibits a consistent selectivity order of $\text{Cd}^{2+} > \text{Cu}^{2+} > \text{Zn}^{2+}$ in competitive ternary metal ion systems simultaneously removing more than 65.51% of Cd^{2+} , 45.57% of Cu^{2+} , and up to 45.44% of Zn^{2+} at RSM-optimized conditions of contact time 28.79 min, pH 9.17, and temperature 40.62 °C, confirming PG as a low-cost adsorbent particularly suited for cadmium removal from contaminated aqueous media. Competitive adsorption behavior differed from single-metal systems, indicating that the presence of multiple metal ions alters total adsorption efficiency and that PG performance in real multi-metal contaminated environments must be assessed under site-specific conditions (75).

Effect of phosphogypsum on crop productivity

The high solubility of Ca^{2+} and SO_4^{2-} from PG, its capacity to improve subsoil chemistry, and its dual role as a calcium and sulfur source have prompted extensive evaluation of PG across cereal, industrial, vegetable, and forage cropping systems in both tropical and temperate agroecosystems (20, 76). Crop response to PG is strongly site-specific and depends on soil chemistry, application rate, crop type, and integration with complementary amendments,

meaning that PG delivers its greatest agronomic benefits where subsoil chemical constraints, particularly aluminum toxicity, sulfur deficiency, or sodium excess, directly limit root growth and nutrient uptake (Table 4). PG consistently promotes root elongation into subsoil layers in acid and sodic cereal systems by alleviating aluminum toxicity and improving Ca^{2+} supply in the rhizosphere, effects that are not achievable through surface lime application alone because lime's neutralizing action is confined to the topsoil while PG's Ca^{2+} and SO_4^{2-} move downward with percolating water (81). In barley grown on saline soils, PG at 15 and 30 t ha⁻¹ enhanced grain yield by 61% and 59%, respectively, compared to the control, with shoot and grain nutrient uptake for P, S, K, and Mg significantly improved and PG outperformed mined gypsum at an equivalent 15 t ha⁻¹ rate, with grain yield 11% and S uptake 64% greater under PG, confirming PG's superior solubility advantage in saline cereal systems (20). Medium- to long-term field trials with surface-applied PG in tropical and subtropical no-till systems have demonstrated consistent grain yield gains of 10%–15% in maize, triticale, rice, and wheat, attributable to increased Ca^{2+} and SO_4^{2-} availability in deeper soil layers, reduced aluminum toxicity, and greater root proliferation into the subsoil (81, 82). Long-term surface application of PG (10 t ha⁻¹) in a no-till Oxisol alleviated drought stress effects on maize by improving soil fertility, root growth, net CO_2 assimilation, and antioxidant enzyme activities compared to the unamended control, with PG combined with lime producing the highest grain yield, demonstrating that PG's agronomic benefits in tropical cereals extend beyond nutrient supply to include physiological stress mitigation. In temperate no-till cereal systems where sulfur deficiency and subsoil acidity are not primary constraints, PG application increased Ca^{2+} and the $\text{Ca}^{2+}/\text{Mg}^{2+}$ ratio in soil layers while reducing Mg^{2+} without significantly affecting grain yield of maize or barley, confirming that yield response to PG in temperate systems is contingent on the presence of specific soil chemical limitations (83).

Among industrial crops, cotton has shown the strongest and most consistent yield response to PG application. Vyshpolsky et al. (46) reported that in magnesium-affected and sodic soils of Central Asia and Kazakhstan, PG application nearly doubled cotton yield in the first year, with yields remaining 70%–85% higher than the control for several subsequent seasons; improvements were attributed to improved soil structure, reduced exchangeable Mg^{2+} and Na^+ , enhanced water infiltration, and improved calcium balance in the root zone with pre-winter application producing the best yield and economic returns. In long-term cotton and sugar beet rotations, repeated PG applications maintained soil chemical balance and sustained productive capacity across multiple seasons, though direct yield measurements were not consistently reported for sugar beet (46). Soybean and common bean exhibited variable yield responses to PG, with gains of 8%–10% reported under no-tillage systems in acid tropical soils; however, PG was consistently less effective than lime at correcting surface acidity in these systems, as PG does not significantly alter soil pH, and this limits its capacity to address pH-driven yield constraints in soils where surface aluminum saturation rather than subsoil acidity or sulfur deficiency is the primary limiting factor (62). Vegetable crop responses to PG are highly variable and strongly rate-dependent. Moderate PG rates improved growth and yield of potato, melon, garlic, cabbage, tomato, and leafy

TABLE 4 Table summarizing the effect of PG application on crop yield and productivity across cereal, vegetable, and fodder crops under field and controlled experimental conditions, organized by crop category.

S. no.	Crop	PG rate (t ha ⁻¹)	Yield/Productivity impact (numeric, from original paper)	Reference
Cereal crops				
2	Wheat (<i>Triticum aestivum</i>), Salt-affected irrigated land, Egypt	~10.39 t ha ⁻¹ + vermicompost	Grain yield and N uptake significantly increased ($p < 0.05$) vs. control. ECE and BS% reduced; CEC and porosity increased.	(28)
3	Wheat (<i>Triticum aestivum</i>), Typic Hapludox, no-till, Brazil	3–12 t ha ⁻¹	New row: Yield increased linearly with PG rate in drought season. S-SO ₄ ²⁻ increased to 0.8 m depth. Mg ²⁺ migrated downward at all rates. Confirmed from Vicensi 2016 Tables 2–4.	(64)
4	Barley (<i>Hordeum vulgare</i>), Calcareous, arid Morocco, field 2 years	1.5, 3, and 4.5 t ha ⁻¹	Grain yield ↑ 21% (1.5 t ha ⁻¹), ↑ 34% (3 t ha ⁻¹), ↑ 39% (4.5 t ha ⁻¹) vs. control. Spikes, tillers, grains per spike, and thousand-grain weight all significantly increased. No heavy metal accumulation effect confirmed.	(51)
5	Barley (<i>Hordeum aestivum</i>), Calcareous soil, pot experiment, Egypt	4.20 g kg ⁻¹ + poultry manure	Barley grain and straw yield significantly increased vs. control. NPK availability, MBC, and soil fauna diversity (Shannon index +19–28%) enhanced. Best: combined PG + PM highest dose.	(54)
6	Maize (<i>Zea mays</i>), Saline-sodic soil, Egypt	5 t ha ⁻¹ + nano-silica	Grain yield ↑ 43.3% vs. control. Improved biomass and water-use efficiency. PG + nano-silica combination more effective than PG alone.	(77)
7	Maize (<i>Zea mays</i>), Saline calcareous, greenhouse, Saudi Arabia	10–50 g kg ⁻¹ + cow manure + inoculants	Plant height, shoot and root dry weight significantly increased. Best: PG 50 g kg ⁻¹ + cow manure + co-inoculation. N, P, K, Fe, Mn, Zn, and Cu uptake all improved.	(69)
8	Rice (<i>Oryza sativa</i>), Saline-sodic soils, NE China	10 t ha ⁻¹	Grain and straw yield significantly increased. PG improved soil structure via Ca ²⁺ –Na ⁺ exchange; soluble Ca ²⁺ increased; ESP and SAR reduced progressively.	(71)
9	Corn (<i>Zea mays</i>), Typic Hapludox, no-till, Brazil	3–12 t ha ⁻¹	New row: Yield response quadratic maximum technical efficiency at 6.38 Mg ha ⁻¹ . Mg ²⁺ migrated downward regardless of rate or splitting (consistent limitation). Soybean yield not affected at any rate.	(64)
Vegetable crops				
10	Onion (<i>Allium cepa</i>), Field trial, Korea	104–312 kg 10 a ⁻¹ (4 levels)	Significant change in soil pH following 7-month onion cultivation with PG. Ca and S content improved.	(78)
11	Garlic (<i>Allium sativum</i>), Field trial, Korea	200 kg 10 a ⁻¹ (optimal rate)	Best effect at 200 kg/10 a. S-related amino acids (cysteine and methionine) in garlic increased. Soil fertility improved.	(78)
12	Chinese cabbage (<i>Brassica oleracea</i>) Cd-contaminated soil, China	Variable + biochar	Leaf biomass productivity increased. Cd immobilization improved. pH buffered with combined PG + biochar. Plant Cd uptake reduced vs. control.	(79)
13	Okra (<i>Abelmoschus esculentus</i>), Alkaline soil, Egypt	Integrated organic + inorganic	Fruit yield per plant and chlorophyll concentration significantly increased. Improved plant productivity under alkaline soil conditions from integrated amendment.	(80)
14	Tomato (<i>Solanum lycopersicum</i>), Saline and agricultural soils, Tunisia	Variable rates	New row: Growth, productivity, and antioxidant enzyme (SOD, CAT, and POD) activities improved at low–moderate PG rates. Excessive rates reduced biomass in agricultural soil. Saline soil showed better response than agricultural soil.	(66)
Fodder/industrial crops				
15	Maize fodder (<i>Zea mays</i>), Saline-sodic soil, Egypt	5 t ha ⁻¹ + nano-silica	Biomass and water-use efficiency significantly increased. Combined PG + nano-silica further improved productivity under saline-sodic conditions.	(77)
16	Buckwheat (<i>Fagopyrum esculentum</i>), Modified PG artificial soil, China	HLRT and ILRT ratios	New row: Yield ↑ 130.6% under HLRT vs. control. BCF for Cd, Cr, Cu, and Zn all <0.3 in grain. Pb from background soil exceeded national food standards. Elevated salinity not yield-limiting under HLRT.	(19)

Yield responses are expressed as percentage change relative to untreated control and are drawn directly from original papers and reflect the strong crop and rate dependence of PG agronomic performance.

vegetables through enhanced calcium and sulfur nutrition, improved soil structure, and increased water use efficiency (66, 67, 78, 79). In potato, PG-amended composts increased tuber yield by over 50% and shifted size distribution toward market-preferred grades. In tomato, PG application at rates not exceeding a 20% pot amendment equivalent improved photosynthetic activity and productivity while lowering soil salinity in saline soils, though this rate must be understood as a greenhouse pot experiment parameter and is not directly equivalent to field application rates (66).

Collectively, PG delivers its greatest and most consistent crop productivity benefits in systems constrained by subsoil aluminum toxicity, sulfur deficiency, and sodium excess conditions prevalent in acid tropical soils, sodic soils of arid and semi-arid regions, and severely saline paddy systems. In temperate and non-constrained systems, yield responses are modest or absent, and in horticultural systems, application rates must be carefully controlled to prevent cadmium accumulation in edible tissues. The evidence across cereal, industrial, and vegetable crops consistently points to the same

conclusion: PG is an effective and targeted soil amendment when matched to the correct soil constraint and applied within safe rate limits (Tables 4, 5).

Risk–benefit analysis and knowledge gap

PG has well-documented agronomic and industrial uses that can add value while reducing waste stockpiles. Applied to soils, PG improves physical structure, assists in sodic soil reclamation, alleviates subsoil acidity, and supplies Ca and S nutrients, benefits consistently reported across field and pot studies, particularly where these elements are limiting (84). On the industrial side, PG can be processed into gypsum board, cement additives, road base, and other construction materials, offering practical routes to valorize large PG volumes (10, 85). Furthermore, the recovery of rare earth elements (REEs) from PG presents a growing opportunity: a probabilistic sustainability assessment combining life cycle and techno-economic analysis showed that bioinspired adsorptive REE separation from PG achieved an internal rate of return above 15% in 87% of simulations, outperforming conventional REE mining in ecosystem quality and resource depletion, though human health impacts require

further attention (86). Substituting PG into goods or land applications reduces environmental impacts associated with big open stacks while also creating economic value in areas with limited natural gypsum sources. Integrating PG into a circular economy framework through both construction material incorporation and agricultural reuse can significantly reduce the effects of traditional stockpiling and landfilling, which currently contribute to soil and water contamination as well as human health risks (10). Despite these advantages, global recycling and reuse of PG is a modest portion of overall output, and growing reuse requires rigorous risk management.

Risks associated with PG depend strongly on its composition, the application rate, and local pathways to soil, water, and biota. PG contains elevated levels of naturally occurring radionuclides, notably ^{226}Ra derived from phosphate rock, and its incorporation into soils can raise soil activity concentrations, making radiological assessment a prerequisite before any application (87). Beyond radioactivity, PG carries chemical impurities and potentially toxic elements. Maazoun et al. (88) showed that while Ni, Cu, and Pb levels in PG-amended soils remained below international standards, Cd and Cr concentrations exceeded Chinese, FAO/WHO, and European allowable limits, with heavy metal bioconcentration factors in plant tissues, particularly roots, increasing progressively as amendment rates rose from 10% to 30%. Human health risk assessment placed children at the highest risk, followed by

TABLE 5 Table documenting the combined effect of PG-based organo-mineral treatments including co-application with organic manure, modified biochar, and microbial inoculants on crop yield and soil chemical properties.

S. no.	Crop/System	PG-OMF combination	Yield impact (from original paper)	Soil property impact (from original paper)	Reference
1	Wheat (<i>Triticum aestivum</i>), Sandy loam, Egypt	PG 2.5 t ha ⁻¹ + modified rice-straw or cotton-stalk biochar 2.5 kg m ⁻²	Grain yield 2.72–2.92 Mg ha ⁻¹ ; straw yield >50% above NPK control. Best at PG + modified biochar (T7–T10 treatments).	SOC +99%; available N + 161%; available P + 102%; available K + 70%; CEC significantly increased. Sandy loam soil quality substantially improved.	(29)
2	Barley (<i>Hordeum aestivum</i>), Calcareous soil, pot, Egypt	PG 4.20 t ha ⁻¹ + poultry manure 4.20–6.30 g kg ⁻¹	Grain and straw yield significantly increased. Best: combined PG + PM highest dose. Soil fauna diversity (Shannon index) +19%–28%.	Available N, P, K, and OM increased; MBC improved; soil pH reduced from 8.08 to 7.64 in calcareous soil.	(54)
3	Maize (<i>Zea mays</i>), Saline calcareous greenhouse, Saudi Arabia	PG 1–5 t ha ⁻¹ + cow manure 50 g kg ⁻¹ + <i>A. chroococcum</i> + <i>B. megaterium</i> + <i>P. fluorescens</i>	Plant height, shoot and root dry weight significantly increased. Best: PG 5 t ha ⁻¹ + cow manure + co-inoculation. N, P, K, Fe, Mn, Zn, and Cu uptake improved.	Soil pH decreased; ECe decreased with inoculation; available P increased significantly; Cd and Pb bioavailability reduced.	(69)
4	Multi-crop rotation (soy–oat–wheat–sorghum), Typic Hapludox, no-till, Brazil (15 years)	PG 2.1 t ha ⁻¹ + lime, long-term	Consistent yield improvement across all rotation crops over 15 years. All four crops (soybean, oat, wheat, sorghum) benefited from lime + PG vs. lime alone.	Ca ²⁺ mobility enhanced to subsoil without surface pH alteration; Al toxicity reduced in subsoil; soil chemical improvement sustained over 15 years.	(62)
5	Maize (<i>Zea mays</i>), Saline-sodic soil, Egypt (2 field seasons)	PG 9 t ha ⁻¹ + <i>A. lipoferum</i> + <i>B. circulans</i> (PGPR)	Chlorophyll, carotenoid, antioxidant enzyme, microbial activity, nutrient absorption, and maize productivity all significantly improved vs. control and vs. PG alone. Best treatment over 2 seasons.	Soil pH, EC, and ESP reduced; urease and dehydrogenase activity improved; microbial communities increased. PG + PGPR combined superior to either alone.	(27)
6	Buckwheat (<i>Fagopyrum esculentum</i>), Reclaimed land, China	Modified PG + red mud artificial soil (HLRT)	New row: Yield +130.6% vs. unmodified control. Grain BCF for Cd, Cr, Cu, and Zn all <0.3. Pb exceeded national food standards due to background soil contamination not attributable to PG.	Bulk density reduced 3.8%–6.9%; moisture +22%; phosphorus mineralization (S-NP) 7.53 U/g; salinity elevated but not yield-limiting under HLRT.	(19)

PG paired with organic materials actively improves nutrient retention, microbial carbon, and yield. PG combinations can enhance aggregate stability and organic matter. Notable yield and soil fertility enhancements arise when PG is integrated with biochar or manure.

These combinations consistently outperform PG applied alone, confirming that integrating PG within a broader soil health management strategy amplifies both agronomic and reclamation outcomes.

adolescents and then adults, leading to the conclusion that PG amendment must not exceed 10% to stay within acceptable sanitary limits (89). Furthermore, Guo et al. (90) confirmed from a broader soil PTE source-apportionment study that agricultural activities are the principal contributor to both non-carcinogenic and carcinogenic health risks in children and adults alike, reinforcing the need for strict dose controls in agricultural PG applications. At higher application rates, PG can also increase total salts and sulfate in soils, alter pH, and affect microbial communities and plant performance in ways that vary considerably by soil type (84, 88). PG stacks generate contaminated leachate, and accurate prediction of leachate volume is a prerequisite for efficient leachate management (91); while controlled soil application can reduce ecological risk relative to unmanaged stacks, aquatic exposure pathways still require evaluation. While ^{226}Ra receives the most regulatory attention, ^{210}Po and ^{210}Pb , both decay products of ^{226}Ra in the uranium-238 decay series, can accumulate in plant tissues through root uptake and represent an undercharacterized food chain transfer risk that current PG agricultural studies rarely measure or report (6). Equal analytical weight must therefore be given to radionuclide and heavy metal characterization as to agronomic benefit, as source-specific compositional assessment of ^{226}Ra , ^{210}Po , ^{210}Pb , Cd, Cr, and As is a non-negotiable prerequisite before any PG batch is recommended for food crop production (10).

Prospects and perspectives

The cooperative association of national and international regulatory organizations, such as FAO and IAEA, must develop some recommendations to stop the widespread adoption of PG that is agro-ecologically unstable, does not adhere to the circular economy aims, and avoids regulatory fragmentation. Other important aspects we need to work on are developing an evaluation of PG performance based on the execution of multi-year, multi-site field trials on different cropping systems (cereals, legumes, oilseeds, and vegetables), with diverse soil types (including Oxisols, Vertisols, Mollisols, and saline-sodic soils) and across different agro-climatic zones. These trials must focus on a deep analysis and comprehensive investigation for monitoring soil health parameters, including radionuclide migration, heavy metal speciation, microbial community dynamics, aggregate stability, and crop quality parameters for at least 5 years. Future PG-based research should focus on developing soil-resilient microbiomes through the co-application of PG with plant growth-promoting rhizobacteria (PGPR), arbuscular mycorrhizal fungi (AMF), and BC, which would be helpful in mitigating the dose-dependent microbial toxicity while maximizing the nutrient availability benefits of PG. Moreover, microbial community structure composition and functional dynamics for PG-amended soil or PG-amended with other organic amendments has not been largely investigated and explored. Metagenomics, metatranscriptomics, and network co-occurrence analyses of PG-amended soil microbiomes can provide the best solution for understanding the functional dynamics of the microbial communities and their benefits for plant growth and crop productivity.

We must also explore strategies to optimize the PG usage efficiencies while reducing their environmental risks with precision agriculture technologies, including remote sensing-based soil health mapping, variable-rate application machinery, and AI-driven decision support systems. The safe agronomic use envelope of PG could be significantly expanded beyond the single-rate applications that predominate in the existing literature by using site-specific treatment rates calibrated to local soil chemistry, crop sensitivity, and proximity to water bodies. In arid and semi-arid regions, which are currently among the most food-insecure in the world, soil degradation from salinization, desertification, and erosion is predicted to worsen under anticipated climate change scenarios.

Conclusion

PG has been proven to be an effective and versatile soil amendment with significant agronomic value for various degraded soil environments. The evidence shows that PG enhances aggregate stability, decreases bulk density, and increases porosity and water infiltration and permeability, especially in sandy, compacted, and sodic soils. Because of its high solubility, it can quickly deliver calcium to enhance flocculation of dispersed clays and alleviate surface sealing and erosion. In saline, sodic, and magnesium-affected soils, PG can exchange Ca^{2+} - Na^+ and Ca^{2+} - Mg^{2+} to decrease ESP, SAR, and EMP, thus restoring soil structure and hydraulic conductivity. As a source of nutrients, it provides available Ca and sulfate-sulfur, which are required for many physiological and metabolic processes, promote root development, and increase crop productivity.

Author contributions

RZ: Conceptualization, Supervision, Writing – review & editing, Validation, Software, Visualization, Project administration, Funding acquisition. IT: Supervision, Conceptualization, Project administration, Visualization, Funding acquisition, Validation, Resources, Writing – review & editing, Writing – original draft. SB: Visualization, Funding acquisition, Project administration, Validation, Formal analysis, Writing – review & editing, Writing – original draft. OK: Validation, Funding acquisition, Supervision, Conceptualization, Writing – review & editing, Writing – original draft, Software, Visualization, Formal analysis. MA: Writing – review & editing, Visualization, Software, Writing – original draft. WA: Writing – original draft, Validation, Software, Writing – review & editing, Conceptualization.

Funding

The author(s) declared that financial support was received for this work and/or its publication. This research was funded by the non-profit joint-stock company “Korkyt Ata Kyzylorda University” under contract No. 121-01-26 dated January 9, 2026.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that generative AI was not used in the creation of this manuscript.

Any alternative text (alt text) provided alongside figures in this article has been generated by Frontiers with the support of artificial

intelligence and reasonable efforts have been made to ensure accuracy, including review by the authors wherever possible. If you identify any issues, please contact us.

Publisher's note

All claims expressed in this article are solely those of the authors and do not necessarily represent those of their affiliated organizations, or those of the publisher, the editors and the reviewers. Any product that may be evaluated in this article, or claim that may be made by its manufacturer, is not guaranteed or endorsed by the publisher.

References

- Heckenmüller M, Narita D, Klepper G. Global availability of phosphorus and its implications for global food supply: an economic overview. In: *Kiel working paper no. 1897*. Kiel: Kiel Institute for the World Economy (IfW). (2014).
- Kirchgesser J, Hazarika M, Bachmann-Pfabe S, Dehmer KJ, Kavka M, Uptmoor R. Phenotypic variation of root-system architecture under high P and low P conditions in potato (*Solanum tuberosum* L.). *BMC Plant Biol.* (2023) 23:68. doi: 10.1186/s12870-023-04070-9
- Mogollón JM, Bouwman AF, Beusen AH, Lassaletta L, van Grinsven HJ, Westhoek H. More efficient phosphorus use can avoid cropland expansion. *Nat Food.* (2021) 2:509–18. doi: 10.1038/s43016-021-00303-y
- Roberts TL, Johnston AE. Phosphorus use efficiency and management in agriculture. *Resour Conserv Recycl.* (2015) 105:275–81. doi: 10.1016/j.resconrec.2015.09.013
- Preston CL, Ruiz Diaz DA, Mengel DB. Corn response to long-term phosphorus fertilizer application rate and placement with strip-tillage. *Agron J.* (2019) 111:841–50. doi: 10.2134/agronj2017.07.0422
- Guillén J, Salas A, Muñoz-Muñoz JG. Assessment of the radiological environmental impact of using phosphogypsum as soil amendment: soil availability modification and radiological impact evaluation. *Chemosphere.* (2025) 387:144647. doi: 10.1016/j.chemosphere.2025.144647
- Yang L, Zhang Y, Yan Y. Utilization of original phosphogypsum as raw material for the preparation of self-leveling mortar. *J Clean Prod.* (2016) 127:204–13. doi: 10.1016/j.jclepro.2016.04.054
- Khalil F, Pagnanelli F, Moscardini E. Phosphogypsum as the secondary source of rare earth elements. *Sustainability.* (2025) 17:1–40. doi: 10.3390/su17198828
- Bilal E, Bellefqih H, Bourgier V, Mazouz H, Dumitras DG, Bard F, et al. Phosphogypsum circular economy considerations: a critical review from more than 65 storage sites worldwide. *J Clean Prod.* (2023) 414:137561. doi: 10.1016/j.jclepro.2023.137561
- Maina L, Kiegiel K, Zakrzewska-Koltuniewicz G. Challenges and strategies for the sustainable environmental management of phosphogypsum. *Sustainability.* (2025) 17:3473. doi: 10.3390/su17083473
- Tayibi H, Choura M, López FA, Alguacil FJ, López-Delgado A. Environmental impact and management of phosphogypsum. *J Environ Manage.* (2009) 90:2377–86. doi: 10.1016/j.jenvman.2009.03.007
- Hu X, Wang J, Wu F, Li D, Yang J, Chen J, et al. Phosphorus recovery and resource utilization from phosphogypsum leachate via membrane-triggered adsorption and struvite crystallization approach. *Chem Eng J.* (2023) 471:144310. doi: 10.1016/j.cej.2023.144310
- Xie Y, Wu Z, Xie X, Fu S, Liu S, Mou S, et al. Simplification of the pretreatment method for phosphate oxygen isotope measurement in phosphogypsum leachate. *J Environ Manage.* (2024) 351:119869. doi: 10.1016/j.jenvman.2023.119869
- Liu Y, Zhang L, Chen L, Xue B, Wang G, Zhu G, et al. Potential of artificial soil preparation for vegetation restoration using red mud and phosphogypsum. *Sci Total Environ.* (2024) 941:173553. doi: 10.2139/ssrn.4728557
- Qin X, Cao Y, Guan H, Hu Q, Liu Z, Xu J, et al. Resource utilization and development of phosphogypsum-based materials in civil engineering. *J Clean Prod.* (2023) 387:135858. doi: 10.1016/j.jclepro.2023.135858
- Kadirova ZC, Hojamberdiev M, Bo L, Hojiyev R, Okada K. Ion uptake properties of low-cost inorganic sorption materials in the CaO–Al₂O₃–SiO₂ system prepared from phosphogypsum and kaolin. *J Clean Prod.* (2014) 83:483–90. doi: 10.1016/j.jclepro.2014.06.084
- Rashad AM. Potential use of phosphogypsum in alkali-activated fly ash under the effects of elevated temperatures and thermal shock cycles. *J Clean Prod.* (2015) 87:717–25. doi: 10.1016/j.jclepro.2014.09.080
- Hossain A, Krupnik TJ, Timsina J, Mahboob MG, Chaki AK, Farooq M, et al. Agricultural land degradation: processes and problems undermining future food security. In: *Environment, climate, plant and vegetation growth*. Springer International Publishing, Cham (2020). p. 17–61.
- Gao L, Li R, Yang D, Bao L, Zhang N. Phosphogypsum improves soil and benefits crop growth: an effective measure for utilizing solid waste resources. *Sci Rep.* (2025) 15:11827. doi: 10.1038/s41598-025-97216-8
- Akfas F, Elghali A, Aboulaich A, Munoz M, Benzaazoua M, Bodinier JL. Exploring the potential reuse of phosphogypsum: a waste or a resource? *Sci Total Environ.* (2024) 908:168196. doi: 10.1016/j.scitotenv.2023.168196
- Du MS, Long B, Mei YZ, Li YS, Li S, Wu XQ, et al. Toward zero-waste phosphogypsum valorization: reengineered reverse-direct flotation synchronizes gypsum purification and functional co-products production. *Chem Eng J.* (2025) 517:164341. doi: 10.1016/j.cej.2025.164341
- Mahmoud E, Ghoneim AM, El Baroudy A, Abd El-Kader N, Aldhumri SA, Othman S, et al. Effects of phosphogypsum and water treatment residual application on key chemical and biological properties of clay soil and maize yield. *Soil Use Manag.* (2021) 37:494–503. doi: 10.1111/sum.12583
- Akanova NI, Vizirskaya MM, Seregin MB, Grebennikova TV. The neutralized phosphogypsum as gypsum-containing meliorant: Russian case-study. *Int Agric J.* (2019) 2(2):12–8. doi: 10.24411/2588-0209-2019-10048
- Zhang B, Chen K, Zhang K, Luo D. Mechanical properties and modification mechanism of phosphogypsum stabilized soil. *SN Appl Sci.* (2022) 4:241. doi: 10.1007/s42452-022-05124-7
- OutbakatMB, Choukr-AllahR, ElGharousM, ElOmariK, SoulaïmaniA, ElMejadedK. Does phosphogypsum application affect salts, nutrients, and trace elements displacement from saline soils?. *Frontiers in Environmental Science.* (2022) 10:964698. doi: 10.3389/fenvs.2022.964698
- Alva AK, Sumner ME. Amelioration of acid soil infertility by phosphogypsum. *Plant and Soil.* (1992) 142:257–64. doi: 10.1007/BF00010970
- Khalifa TH, Elbagory M, Omara AED. Salt stress amelioration in maize plants through phosphogypsum application and bacterial inoculation. *Plants.* (2021) 10:2024. doi: 10.3390/plants10102024
- Rashed S, Abo-Elela E, Amer M. Improving and sustaining some soil properties by application of some soil amendments and N-fertilizer and its effect on productivity of wheat and rice crops under salt-affected soils. *Asian J Soil Sci Plant Nutr.* (2022) 8:1–12. doi: 10.9734/ajsspn/2022/v8i230134
- Elbagory M, Shaker EM, El-Nahrawy S, Omara AED, Khalifa TH. The concurrent application of phosphogypsum and modified biochar as soil amendments influence sandy soil quality and wheat productivity. *Plants.* (2024) 13:1492. doi: 10.3390/plants13111492
- Materchera SA. Aggregation in a surface layer of a hardsetting and crusting soil as influenced by the application of amendments and grass mulch in a South African semi-arid environment. *Soil Tillage Res.* (2009) 105:251–9. doi: 10.1016/j.still.2009.07.008
- Makarova TK, Maksymova NN, Hapich GV, Chushkina IV. Redistribution of particle-size fractions in ordinary chernozem affected by long-term irrigation and chemical melioration with phosphogypsum. *Land Reclam Water Manag.* (2020), 95–101. doi: 10.31073/mivg202001-238

32. Onoprienko DM, Makarova TK, Tkachuk AV, Hapich HV, Roubik H. The influence of phosphogypsum on the salt composition of salinated soil. *Land Reclam Water Manag.* (2023) 1:102–13. doi: 10.31073/mirg202301-350
33. Wang WH, Li JB, Zhang ZP, Wang QJ. Effects of soil amendments on soil air permeability in film mulched drip irrigation field. *Trans Chin Soc Agric Mach.* (2015) 46:160–7. doi: 10.6041/j.issn.1000-1298.2015.06.049
34. Vyshpolsky F, Mukhamedjanov K, Bekbaev U, Ibatullin S, Yuldashev T, Noble AD, et al. Optimizing the rate and timing of phosphogypsum application to magnesium-affected soils for crop yield and water productivity enhancement. *Agric Water Manag.* (2010) 97:1277–86. doi: 10.1016/j.agwat.2010.02.020
35. Gharaibeh MA, Rusan MJ, Eltaif NI, Shunnar OF. Reclamation of highly calcareous saline-sodic soil using low quality water and phosphogypsum. *Appl Water Sci.* (2014) 4:223–30. doi: 10.1007/s13201-014-0189-3
36. BriedisC, SáJCM, CairesEF, NavarroJF, InagakiTM, BoerA, et al. Soil organic matter pools and carbon-protection mechanisms in aggregate classes influenced by surface liming in a no-till system. *Geoderma.* (2012) 170:80–88. 10.1016/j.geoderma.2011.10.011.
37. Alves LA, Fontoura SMV, Tiecher T, Pesini G, Flores JPM, Filippi D, et al. Long-term soil acidity dynamics and crop yield response to phosphogypsum and limestone with different reactivities in a no-till Oxisol. *Eur J Agron.* (2025) 171:127803. doi: 10.1016/j.eja.2025.127803
38. Hasana H, Beyene S, Kifilu A, Kidanu S. Effect of phosphogypsum amendment on chemical properties of sodic soils at different incubation periods. *Appl Environ Soil Sci.* (2022) 2022:9097994. doi: 10.1155/2022/9097994
39. Chen W, Luo Z, Sun T, Tang P, Jin D. Utilization of high-volume phosphogypsum in artificial aggregate by compaction granulation: effects of muck on physical properties, strength and leaching stability. *J Sustain Cem Based Mater.* (2023) 12:951–61. doi: 10.1080/21650373.2022.2141368
40. Liang Y, Guan B, Cao T, Liu G, Tang P, He M, et al. Study on the properties of an excess-sulphate phosphogypsum slag cement stabilized base-course mixture containing phosphogypsum-based artificial aggregate. *Constr Build Mater.* (2023) 409:134095. doi: 10.1016/j.conbuildmat.2023.134095
41. Mamedov AI, Shainberg I, Wagner LE, Warrington DN, Levy GJ. Infiltration and erosion in soils treated with dry PAM, of two molecular weights, and phosphogypsum. *Soil Res.* (2009) 47:788–95. doi: 10.1071/sr09027
42. Tang Z, Lei T, Yu J, Shainberg I, Mamedov AI, Ben-Hur M, et al. Runoff and interrill erosion in sodic soils treated with dry PAM and phosphogypsum. *Soil Sci Soc Am J.* (2006) 70:679–90. doi: 10.2136/sssaj2004.0395
43. Tian Y, Li Z, Wang S, Zhou Y, Li D, Fu L. Application of MICP in water stability and hydraulic erosion control of phosphogypsum material in slope. *Appl Sci.* (2022) 12:1783. doi: 10.3390/app12041783
44. Cochrane BHW, Reichert JM, Eltz FLF, Norton LD. Controlling soil erosion and runoff with polyacrylamide and phosphogypsum on subtropical soil. *Trans ASAE.* (2005) 48:149–54. doi: 10.13031/2013.17958
45. Churka Blum S, Caires EF, Alleoni LR. Lime and phosphogypsum application and sulfate retention in subtropical soils under no-till system. *J Soil Sci Plant Nutr.* (2013) 13:279–300. doi: 10.4067/s0718-95162013005000024
46. Vyshpolsky F, Qadir M, Karimov A, Mukhamedjanov K, Bekbaev U, Paroda R, et al. Enhancing the productivity of high-magnesium soil and water resources in Central Asia through the application of phosphogypsum. *Land Degrad Dev.* (2008) 19:45–56. doi: 10.1007/978-3-319-01017-5_28
47. Onoprienko D, Makarova T, Hapich H, Chernysh Y, Roubik H. Agroecological transformation in the salt composition of soil under the phosphogypsum influence on irrigated lands in Ukraine. *Agriculture.* (2024) 14(3):408. 10.3390/agriculture14030408.
48. Quintero JM, Enamorado S, Mas JL, Abril JM, Polvillo O, Delgado A. Phosphogypsum amendments and irrigation with acidulated water affect tomato nutrition in reclaimed marsh soils from SW Spain. *Spanish Journal of Agricultural Research.* (2014) 12(3):809–19. 10.5424/sjar/2014123-5273.
49. Li L, Yang S, Hu X, LiZ, Chen H. The combined application of salt-alkali tolerant phosphate solubilizing microorganisms and phosphogypsum is an excellent measure for the future improvement of saline-alkali soils. *Frontiers in Microbiology.* (2024) 15:1364487. 10.3389/fmicb.2024.1364487.
50. Kim YN, Cho JY, Yoon YE, Choe HJ, Cheong MS, Lee M, et al. Influences of phosphogypsum application on soil property and yield and quality of onion (*Allium cepa* L.). *Korean J Soil Sci Fertil.* (2021) 54:141–50. doi: 10.7745/kjssf.2021.54.2.141
51. Outbakat MB, Bouray M, Choukr-Allah R, El Gharous M, El Omari K, El Mejahed K. Phosphogypsum as fertilizer: impacts on soil fertility, barley yield components, and heavy metals contents. *Plants.* (2024) 14:16. doi: 10.3390/plants14010016
52. Komissarov M, Gabbasova I, Garipov T, Suleymanov R, Sidorova L. The effect of phosphogypsum and Turkey litter application on the properties of eroded agrochernozem in the south ural region (Russia). *Agronomy.* (2022) 12:2594. doi: 10.3390/agronomy12112594
53. He C, He M, Qu G, Xu R, Wei J, Cai Y, et al. Co-utilization of phosphogypsum and agricultural organic solid waste for production of nutrient-rich soil: an effective approach for solid waste treatment. *J Mater Cycles Waste Manag.* (2026) 28:141–56. doi: 10.1007/s10163-025-02399-x
54. Mahmoud E, Ghoneim AM, Selem M, Zuhair R, El-Refaey A, Khalafallah N. Phosphogypsum and poultry manure enhance diversity of soil fauna, soil fertility, and barley (*Hordeum aestivum* L.) grown in calcareous soils. *Sci Rep.* (2023) 13:9944. doi: 10.1038/s41598-023-37021-3
55. Karpenko NP, Egemberdiev DK, Glazunova IV. Bioremediation for the restoration of saline and degraded soils in the arid zone. *IOP Conf Ser Earth Environ Sci.* (2022) 1010:12044. doi: 10.1088/1755-1315/1010/1/012044
56. Li C, Dong Y, Yi Y, Tian J, Xuan C, Wang Y, et al. Effects of phosphogypsum on enzyme activity and microbial community in acid soil. *Sci Rep.* (2023) 13:6189. doi: 10.1038/s41598-023-33191-2
57. Gabsi H, Tallou A, Aziz F, BoukChina R, Karboub N, Caceres LA, et al. Application of phosphogypsum and organic amendment for bioremediation of degraded soil in Tunisia oasis: targeting circular economy. *Sustainability.* (2023) 15:4769. doi: 10.3390/su15064769
58. Bouray M, Moir J, Condron L, Lehto N. Impacts of phosphogypsum, soluble fertilizer and lime amendment of acid soils on the bioavailability of phosphorus and sulphur under lucerne (*Medicago sativa*). *Plants.* (2020) 9:883. doi: 10.3390/plants9070883
59. Masud MM, Li JY, Xu RK. Application of alkaline slag and phosphogypsum for alleviating soil acidity in an Ultisol profile: a short-term leaching experiment. *J Soils Sediments.* (2015) 15:365–73. doi: 10.1007/s11368-014-0993-2
60. Crusciol CA, Artigiani AC, Arf O, Carmeis Filho AC, Soratto RP, Nascente AS, et al. Soil fertility, plant nutrition, and grain yield of upland rice affected by surface application of lime, silicate, and phosphogypsum in a tropical no-till system. *Catena.* (2016) 137:87–99. doi: 10.1016/j.catena.2015.09.009
61. Wen S, Jiang R, Yang X, Chen X, Xia T, Wu L, et al. *Bacillus velezensis* combined with phosphogypsum promotes the growth of *Cynodon dactylon* L. by influencing rhizosphere soil nutrients, Pb bioavailability and microbial community composition in lead-contaminated soil. *Environ Technol Innov.* (2025) 37:103956. doi: 10.1016/j.eti.2024.103956
62. da Costa CHM, Crusciol CAC. Long-term effects of lime and phosphogypsum application on tropical no-till soybean–oat–sorghum rotation and soil chemical properties. *Eur J Agron.* (2016) 74:119–32. doi: 10.1016/j.eja.2015.12.001
63. Al-Enazy AA, Al-Barakah F, Al-Oud S, Usman A. Effect of phosphogypsum application and bacteria co-inoculation on biochemical properties and nutrient availability to maize plants in a saline soil. *Arch Agron Soil Sci.* (2018) 64:1394–406. doi: 10.1080/03650340.2018.1437909
64. Vicensi M, Müller MML, Kawakami J, Nascimento RD, Michalovicz L, Lopes C. Do rates and splitting of phosphogypsum applications influence the soil and annual crops in a no-tillage system? *Rev Bras Cienc Solo.* (2016) 40:1–17. doi: 10.1590/18069657rbcs20150155
65. Soratto RP, Crusciol CA. Dolomite and phosphogypsum surface application effects on annual crops nutrition and yield. *Agron J.* (2008) 100:261–70. doi: 10.2134/agronj2007.0120
66. Smaoui-Jardak M, Kriaa W, Maalej M, Zouari M, Kamoun L, Trabelsi W, et al. Effect of the phosphogypsum amendment of saline and agricultural soils on growth, productivity and antioxidant enzyme activities of tomato (*Solanum lycopersicum* L.). *Ecotoxicology.* (2017) 26:1089–104. doi: 10.1007/s10646-017-1836-x
67. Takasu E, Yamada F, Shimada N, Kumagai N, Hirabayashi T, Saigusa M. Effect of phosphogypsum application on the chemical properties of Andosols, and the growth and Ca uptake of melon seedlings. *Soil Sci Plant Nutr.* (2006) 52:760–8. doi: 10.1111/j.1747-0765.2006.00093.x
68. Nascimento R, Souza JAD, Moreira A, Moraes LAC. Phosphogypsum and vinasse application: soil chemical properties and alfalfa productivity and nutritional characteristics. *Rev Caatinga.* (2017) 30:213–9. doi: 10.1590/1983-21252017v30n123rc
69. Al-Enazy AAR, Al-Oud SS, Al-Barakah FN, Usman AR. Role of microbial inoculation and industrial by-product phosphogypsum in growth and nutrient uptake of maize (*Zea mays* L.) grown in calcareous soil. *J Sci Food Agric.* (2017) 97:3665–74. doi: 10.1002/jsfa.8226
70. Al-Hwaiti M, Al-Khashman O. Health risk assessment of heavy metals contamination in tomato and green pepper plants grown in soils amended with phosphogypsum waste materials. *Environ Geochem Health.* (2015) 37:287–304. doi: 10.1007/s10653-014-9646-z
71. Liu M, Liang Z, Ma HY, Huang LH, Wang MM. Responses of rice (*Oryza sativa* L.) growth and yield to phosphogypsum amendment in saline-sodic soils of north-east China. *J Food Agric Environ.* (2010) 8:827–33. doi: 10.1002/jsfa.8226
72. Al-Rubaie SH, Al-Hassoon SN, Dwene SJ. Effect of the type and level of organic amendments and phosphogypsum of soil affected by salinity planted with sorghum on some chemical properties of the soil. *J Al-Muthanna Agric Sci.* (2025) 12:1–18. doi: 10.52113/mjas04/12.1/2
73. Nayak AK, Raja R, Rao KS, Shukla AK, Mohanty S, Shahid M, et al. Effect of fly ash application on soil microbial response and heavy metal accumulation in soil and rice plant. *Ecotoxicol Environ Saf.* (2015) 114:257–62. doi: 10.1016/j.ecoenv.2014.03.033
74. Mahmoud E, Abd El-Kader N. Heavy metal immobilization in contaminated soils using phosphogypsum and rice straw compost. *Land Degrad Dev.* (2015) 26:819–24. doi: 10.1002/ldr.2288

75. Es-Said A, Nafai H, El Hamdaoui L, Bouhaouss A, Bchitou R. Adsorptivity and selectivity of heavy metals Cd(II), Cu(II), and Zn(II) toward phosphogypsum. *Desalin Water Treat.* (2020) 197:291–9. doi: 10.5004/dwt.2020.25964
76. Bing L, Jiancheng S, Mengjun C, Xiangfei Z, Renlong L, Yong Y. A new basic burning raw material for simultaneous stabilization/solidification of PO₄-P and F- in phosphogypsum. *Ecotoxicol Environ Saf.* (2023) 252:114582. doi: 10.1016/j.ecoenv.2023.114582
77. Goda DA, El-Gamal EH, Rashad M, Abdel-Fattah YR. The optimization of calcareous soil cation exchange capacity via the feather hydrolysate and NP fertilizers integration. *Sci Rep.* (2025) 15:4676. doi: 10.1038/s41598-025-86941-9
78. Kim YN, Cho JY, Yoon YE, Choe HJ, Cheong MS, Kim KR, et al. Effects of phosphogypsum application on field soil properties and yield and quality of garlic (*Allium sativum* L.). *Korean J Environ Agric.* (2021) 40:33–9. doi: 10.5338/kjea.2021.40.1.4
79. Mu L, Zhou H, Li A, Wang L, Wang J, Sun S, et al. The effects of the combined application of biochar and phosphogypsum on the physicochemical properties of Cd-contaminated soil and the yield quality of Chinese cabbage. *Agriculture.* (2024) 14:1865. doi: 10.3390/agriculture14111865
80. Abd El-Hady MA, Mosaad IS. Integrated organic and inorganic amendments for improving productivity of okra (*Abelmoschus esculentus* L.) in alkaline soil. *Egypt J Soil Sci.* (2024) 64:207–19. doi: 10.21608/ejss.2023.246724.1685
81. Meena RK, Meena R, Ram B. Effect of fly ash, phospho-gypsum and chemical fertilizers on growth, yield and nutrient uptake of wheat (*Triticum aestivum* L.) in inceptisol. *J Pharmacogn Phytochem.* (2019) 8:733–6.
82. Michalovicz L, Müller MML, Tormena CA, Dick WA, Vicensi M, Meert L. Soil chemical attributes, nutrient uptake and yield of no-till crops as affected by phosphogypsum doses and parceling in southern Brazil. *Arch Agron Soil Sci.* (2019) 65:385–99. doi: 10.1080/03650340.2018.1505041
83. Fontoura SMV, de Castro Pias OH, Tiecher T, Cherubin MR, de Moraes RP, Bayer C. Effect of gypsum rates and lime with different reactivity on soil acidity and crop grain yields in a subtropical Oxisol under no-tillage. *Soil Tillage Res.* (2019) 193:27–41. doi: 10.1016/j.still.2019.05.005
84. Yamani B, Hamza A, Yamani A, Batbat A, Zouahri A, Guilli ME, et al. From byproduct to breakthrough: agronomic, environmental, and regulatory aspects of phosphogypsum use in agriculture. *Agronomy.* (2026) 16(4):461. doi: 10.3390/agronomy16040461
85. Men J, Li Y, Cheng P, Zhang Z. Recycling phosphogypsum in road construction materials and associated environmental considerations: a review. *Heliyon.* (2022) 8. doi: 10.1016/j.heliyon.2022.e11518
86. Smerigan A, Shi R. (2024). “Advancing the economic and environmental sustainability of rare earth element recovery from phosphogypsum using a selective bio-based adsorptive separation”, in: *2024 AIChE Annual Meeting Proceedings*. New York: American Institute of Chemical Engineers (AIChE).
87. Papastefanou C, Stoulos S, Ioannidou A, Manolopoulou M. The application of phosphogypsum in agriculture and the radiological impact. *J Environ Radioact.* (2006) 89:188–98. doi: 10.1016/j.jenvrad.2006.05.005
88. Maazoun H, Bouassida M. Phosphogypsum management perspectives: massive valorization or massive storage. *Acta Sci Agric.* (2019) 3:184–9. doi: 10.31080/asag.2019.03.0584
89. Ben Chabchoubi I, Bouguerra S, Ksibi M, Hentati O. Health risk assessment of heavy metals exposure via consumption of crops grown in phosphogypsum-contaminated soils. *Environ Geochem Health.* (2021) 43:1953–81. doi: 10.1007/s10653-020-00777-y
90. Guo G, Wang Y, Zhang D, Lei M. Source-specific ecological and health risks of potentially toxic elements in agricultural soils in Southern Yunnan Province and associated uncertainty analysis. *J Hazard Mater.* (2021) 417:126144. doi: 10.1016/j.jhazmat.2021.126144
91. Meng M, Luo W, Wang S, Zeng G. Predicting the leachate generation from wet phosphogypsum stack using a water-balance-analysis based model. *Environ Res.* (2022) 212:113338. doi: 10.1016/j.envres.2022.113338